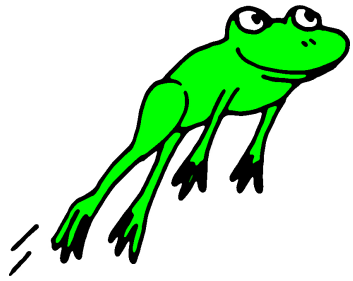
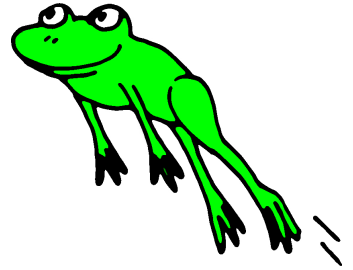


An introduction to deep inference



Victoria Barrett
Inria Saclay, Équipe Partout



Proof Society 2026, Aussois

What is deep inference?

Deep inference is a methodology for the design of proof formalisms.

Originally developed by Alessio Guglielmi to allow for a proof-theoretic treatment of sequential composition.

Emphasises:

- less syntax and bureaucracy,
- underlying geometric nature of proofs,
- regularity of inference rules and normalisation procedures in different systems

What is a proof formalism?

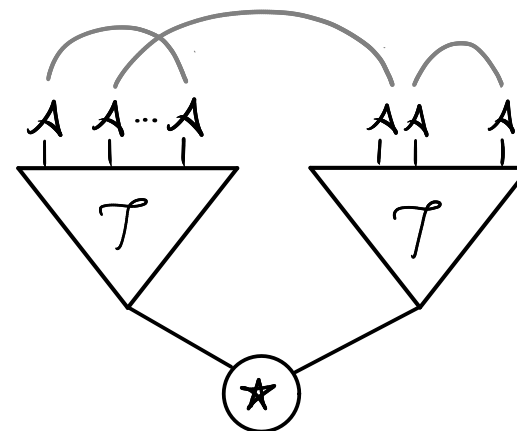
Sequent Calculus

$$\mathcal{F} ::= \mathcal{A} \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

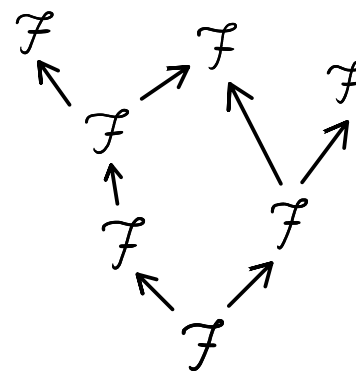
$$\mathcal{S} ::= \mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}$$

$$\triangleleft_D \quad ::= \quad \frac{\triangleleft_D \quad \dots \quad \triangleleft_D}{\mathcal{S}}$$

Proof Nets



Hilbert systems



What is a proof formalism?

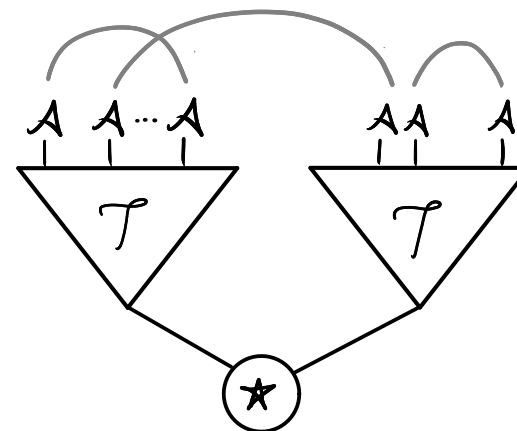
1-sided Sequent Calculus

$$\mathcal{F} ::= \mathcal{A} \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

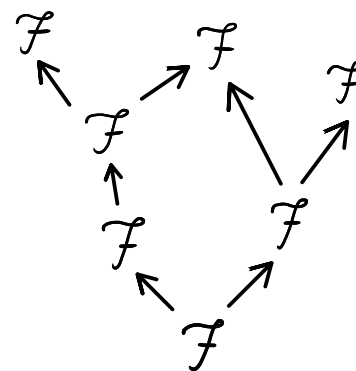
$$\mathcal{S} ::= \cancel{\mathcal{F}, \dots, \mathcal{F}} \vdash \mathcal{F}, \dots, \mathcal{F}$$

$$\triangleleft_D \quad ::= \quad \frac{\triangleleft_D \quad \dots \quad \triangleleft_D}{\mathcal{S}} \quad \mathcal{P}$$

Proof Nets



Hilbert systems



What is a proof formalism?

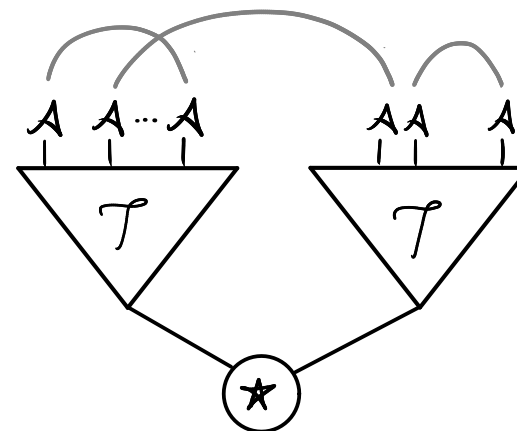
hyper-Sequent Calculus

$$\mathcal{F} ::= \mathcal{A} \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

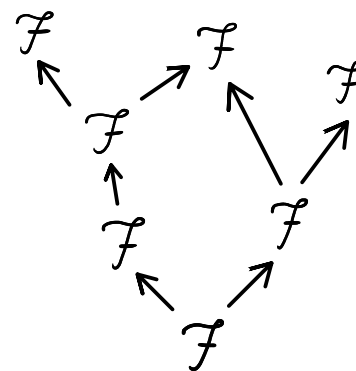
$$\mathcal{S} ::= \mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}$$

$$\triangleleft \mathcal{D} ::= \frac{\triangleleft \mathcal{D} \quad \dots \quad \triangleleft \mathcal{D}}{\mathcal{S} \mid \dots \mid \mathcal{S}}$$

Proof Nets



Hilbert systems



What makes a formalism good?

Normalisation : proofs have useful normal forms and normalisation procedures are simple

Complexity : theorems have small proofs

Semantics : proofs are represented with little syntactic bureaucracy and it's easy to tell when two proofs are the same

Proof search : it's quick to prove or refute formulae

Expressibility : systems can be given for interesting logics

:

Towards open deduction, a deep inference formalism

Sequent Calculus

$$\mathcal{F} ::= A \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

formula-level connectives

$$\triangleleft \mathcal{D} ::= \frac{\triangleleft \mathcal{D} \quad \dots \quad \triangleleft \mathcal{D}}{\mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}}$$

derivation-level connectives

Inference rules have
shallow access to
connectives

For example,

$$\text{contr} \quad \frac{\Gamma \vdash A, A, \Delta}{\Gamma \vdash A, \Delta} \quad \text{not} \quad \frac{\Gamma \vdash A \vee A, \Delta}{\Gamma \vdash A, \Delta}$$

$$\text{and} \quad \wedge \quad \frac{\Gamma \vdash A, \Delta \quad \Gamma \vdash B, \Delta}{\Gamma \vdash A \wedge B, \Delta}$$

$$\text{not} \quad \frac{\Gamma \vdash A \vee C, \Delta \quad \Gamma \vdash B \vee C, \Delta}{\Gamma \vdash (A \wedge B) \vee C, \Delta}$$

Towards open deduction, a deep inference formalism

Sequent Calculus

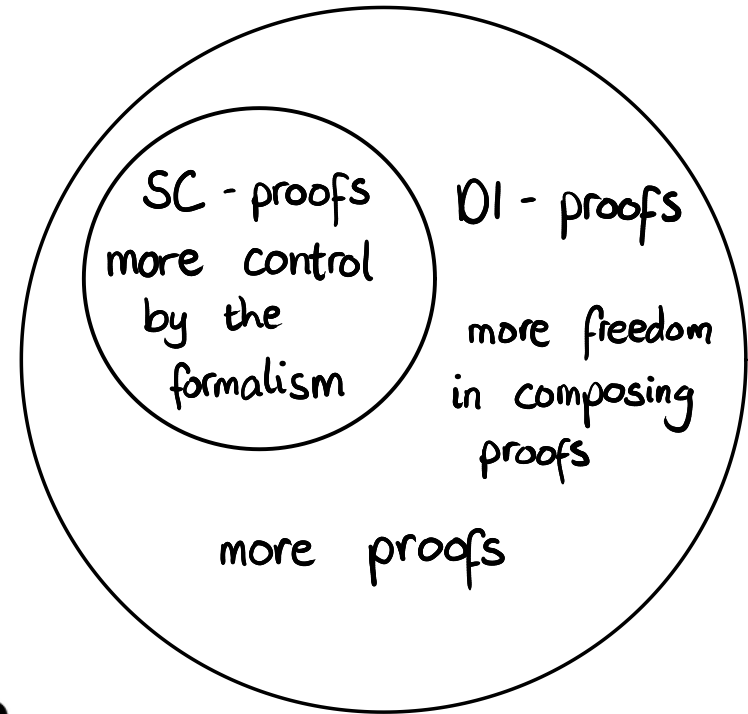
$$\mathcal{F} ::= A \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

formula-level connectives

$$\triangleleft_D ::= \frac{\triangleleft_D \quad \dots \quad \triangleleft_D}{\mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}}$$

derivation-level connectives

Inference rules have
shallow access to
connectives



Towards open deduction, a deep inference formalism

Sequent Calculus

$$\mathcal{F} ::= A \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

formula-level connectives

$$\triangle \mathcal{D} ::= \frac{\triangle \mathcal{D} \quad \dots \quad \triangle \mathcal{D}}{\mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}}$$

derivation-level connectives

Inference rules have
shallow access to
connectives

Open deduction

$$\boxed{\mathcal{D}} ::= A \mid \mathcal{U} \mid \boxed{\mathcal{D}} \odot \boxed{\mathcal{D}} \mid \star \boxed{\mathcal{D}} \mid \frac{\boxed{\mathcal{D}}}{\boxed{\mathcal{D}}}$$

$$\mathcal{F} \subseteq \mathcal{D} \quad \text{one level of connective}$$

Inference rules have deep access
to formulae



Towards open deduction, a deep inference formalism

Sequent Calculus

$$\mathcal{F} ::= A \mid \mathcal{U} \mid \mathcal{F} \odot \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

formula-level connectives

$$\triangle \mathcal{D} ::= \frac{\triangle \mathcal{D} \quad \dots \quad \triangle \mathcal{D}}{\mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}}$$

derivation-level connectives

$$\text{contr} \frac{B, C \vdash A, A, D}{B, C \vdash A, D} \longrightarrow \frac{(B \wedge C) \rightarrow (A \vee A \vee D)}{(B \wedge C) \rightarrow (A \vee D)} \longrightarrow B^\perp \vee C^\perp \vee \frac{A \vee A}{A} \vee D$$

Open deduction

$$\boxed{\mathcal{D}} ::= A \mid \mathcal{U} \mid \boxed{\mathcal{D}} \odot \boxed{\mathcal{D}} \mid \star \boxed{\mathcal{D}} \mid \frac{\boxed{\mathcal{D}}}{\boxed{\mathcal{D}}}$$

$$\mathcal{F} \subseteq \mathcal{D}$$

one level of connective

Towards open deduction, a deep inference formalism

Open deduction

$$\boxed{\mathcal{D}} ::= \mathcal{A} \mid \mathcal{U} \mid \boxed{\mathcal{D}} \odot \boxed{\mathcal{D}} \mid \star \boxed{\mathcal{D}} \mid \text{p} \frac{\boxed{\mathcal{D}}}{\boxed{\mathcal{D}}}$$

$\mathcal{F} \subseteq \mathcal{D}$ one level of connective

$$\frac{\frac{E}{\frac{A}{B} \wedge \frac{C}{D}}}{\frac{F \vee G}{H} \wedge J} \vee \frac{\quad}{K} \xrightarrow[\text{conclusion}]{\text{premise}} \frac{E \vee ((F \vee G) \wedge J)}{(B \wedge D) \vee K}$$

Towards open deduction, a deep inference formalism

Sequent Calculus

$$\mathcal{F} ::= \mathcal{A} \mid \mathcal{U} \mid \mathcal{F} \circ \mathcal{F} \mid \star \mathcal{F} \mid \dots$$

formula-level connectives

$$\triangleleft \mathcal{D} ::= \frac{\triangleleft \mathcal{D} \quad \dots \quad \triangleleft \mathcal{D}}{\mathcal{F}, \dots, \mathcal{F} \vdash \mathcal{F}, \dots, \mathcal{F}}$$

derivation-level connectives

$$\text{cut} \frac{\Gamma \vdash A, \Delta \quad \Xi, A \vdash \Upsilon}{\Gamma, \Xi \vdash \Delta, \Upsilon}$$

Open deduction

$$\boxed{\mathcal{D}} ::= \mathcal{A} \mid \mathcal{U} \mid \boxed{\mathcal{D}} \odot \boxed{\mathcal{D}} \mid \star \boxed{\mathcal{D}} \mid \frac{\boxed{\mathcal{D}}}{\boxed{\mathcal{D}}}$$

$$\mathcal{F} \subseteq \mathcal{D} \quad \text{one level of connective}$$

$$\xrightarrow{s} \frac{\begin{array}{c} \bigwedge_{\Gamma} \mathcal{J} \\ \parallel \\ A \vee V_{\Delta} \mathcal{J} \end{array} \wedge \bigwedge_{\Xi} \mathcal{K}}{V_{\Delta} \mathcal{J} \vee \begin{array}{c} A \wedge \bigwedge_{\Xi} \mathcal{K} \\ \parallel \\ V_{\Gamma} \mathcal{V} \end{array}}$$

Towards open deduction, a deep inference formalism

$$\text{id} \frac{}{\vdash a, a^\perp} \rightarrow \text{id} \frac{1}{a \wp a^\perp}$$

$$\Gamma \vdash \Delta \rightarrow \wp_{\gamma \in \Gamma} \gamma^\perp \wp \wp_{\delta \in \Delta} \delta$$

$$\otimes \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} \rightarrow \begin{array}{c} \text{s} \frac{(A \wp C) \otimes (B \wp D)}{(\text{s} \frac{(A \otimes (B \wp D))}{(A \otimes B) \wp D}) \wp C} \end{array}$$

$$\& \frac{\vdash A, \Gamma \quad \vdash B, \Gamma}{\vdash A \& B, \Gamma} \rightarrow \text{d}\downarrow \frac{(A \wp C) \& (B \wp C)}{(A \& B) \wp c\downarrow \frac{C \oplus C}{C}}$$

We only want inference rules
 $\rho \frac{A}{B}$ that correspond to a provable
 sequent $A \vdash B$



Open deduction : symmetry

$$\text{cut} \frac{\vdash A, \Gamma \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} \rightarrow \frac{\frac{\frac{(A \wp C) \otimes (A^\perp \wp D)}{s} \quad \frac{\frac{(A \wp C) \otimes A^\perp}{s} \wp D}{i\uparrow} \quad \frac{(A \otimes A^\perp) \wp C}{i\uparrow} \quad \perp}{\perp}}{\perp}$$

If $\frac{A}{B}$ in S then $\frac{B^\perp}{A^\perp}$ in $S \cup \{i\downarrow, i\uparrow, s, =\}$

A set S of inference rules is **symmetric** if

for every $\frac{A}{B} \in S$, $\frac{B^\perp}{A^\perp} \in S$

In a symmetric system, derivations can be negated



$$\frac{A}{B} \stackrel{(\oplus)}{=} \frac{B^\perp}{A^\perp}$$

$$\frac{\frac{\frac{B^\perp}{B} \otimes \frac{1}{A \wp A^\perp}}{i\downarrow} \wp A^\perp}{s} = \frac{\perp}{A^\perp}$$

Open deduction : symmetry

$$\textcircled{H} = \frac{\frac{E}{\frac{A}{B} \wedge \frac{C}{D}} \vee \frac{\frac{F \vee G}{H} \wedge J}{K}}$$

$$\textcircled{H}^\perp = \frac{\frac{\frac{B^\perp}{A^\perp} \vee \frac{D^\perp}{C^\perp}}{E^\perp} \wedge \frac{\frac{K^\perp}{H^\perp \vee J^\perp}}{\frac{F^\perp \wedge G^\perp}}}$$

A set S of inference rules is **symmetric** if

for every $\rho \vdash \frac{A}{B} \in S$, $\rho^\perp \vdash \frac{B^\perp}{A^\perp} \in S$



In a symmetric system, derivations
can be negated

Open deduction : atomicity

$$\text{cut} \frac{\frac{}{\vdash A, \Gamma} \quad \frac{}{\vdash A^\perp, \Delta}}{\vdash \Gamma, \Delta} \rightarrow \frac{s \frac{(A \wp C) \otimes (A^\perp \wp D)}{(A \wp C) \otimes A^\perp} \wp D}{i \uparrow \frac{(A \otimes A^\perp) \wp C}{\perp}}$$

The cut rule can be made **atomic**, which is not possible in the sequent calculus

$$d \uparrow \frac{(D \wp E) \otimes (D^\perp \oplus E^\perp)}{i \uparrow \frac{D \otimes D^\perp}{\perp} \oplus i \uparrow \frac{E \otimes E^\perp}{\perp}} \downarrow c \frac{}{\perp}$$

$$= \frac{(D \otimes E) \otimes (D^\perp \wp E^\perp)}{s \frac{E \otimes (D^\perp \wp E^\perp)}{D \otimes D^\perp \wp i \uparrow \frac{E \otimes E^\perp}{\perp}}} = \frac{}{i \uparrow \frac{}{D^\perp} \perp}$$

Open deduction : atomicity

Inference rules of system SMALS_g for multiplicative additive linear logic

identity

$$a \downarrow \frac{1}{a \wp a^\perp}$$

cut

$$a \uparrow \frac{a \otimes a^\perp}{\perp}$$

switch

$$s \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

$$= \frac{A}{B} \quad \text{for } A = B$$

contraction

$$c \downarrow \frac{A \oplus A}{A}$$

cocontraction

$$c \uparrow \frac{A}{A \& A}$$

$$d \downarrow \frac{(A \wp B) \& (C \wp D)}{(A \& C) \wp (B \oplus D)}$$

$$d \uparrow \frac{(A \& B) \otimes (C \oplus D)}{(A \otimes C) \oplus (B \otimes D)}$$

$$t \downarrow \frac{0}{A}$$

$$t \uparrow \frac{A}{\top}$$

such that $\otimes, \wp, \&, \oplus$ are commutative and associative and have units $1, \perp, \top, 0$



Note that there is a difference between our cocontraction and DiLL's cocontraction: theirs is a flipped contraction, ours is a flipped *and dualised* contraction.

Open deduction : atomicity

Inference rules of system SKSg for propositional classical logic

$$\text{identity} \quad \text{ai} \downarrow \frac{\top}{a \vee a^\perp}$$

$$\text{cut} \quad \text{ai} \uparrow \frac{a \wedge a^\perp}{\perp}$$

$$\text{switch} \quad \text{s} \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C}$$

$$= \frac{A}{B} \quad \text{for } A = B$$

$$\text{contraction} \quad \text{c} \downarrow \frac{A \vee A}{A}$$

$$\text{cocontraction} \quad \text{c} \uparrow \frac{A}{A \wedge A}$$

$$\text{weakening} \quad \text{w} \downarrow \frac{\perp}{A}$$

$$\text{coweakening} \quad \text{w} \uparrow \frac{A}{\top}$$

such that $\wedge, \vee$
are commutative and
associative and have
units $\top, \perp$,

Open deduction : atomicity

Inference rules of system SKSg for propositional classical logic

$$\text{identity} \quad \text{ai} \downarrow \frac{\top}{a \vee a^\perp}$$

$$\text{cut} \quad \text{ai} \uparrow \frac{a \wedge a^\perp}{\perp}$$

$$\text{switch} \quad \text{s} \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C} = \frac{A}{B}$$

$$\text{contraction} \quad \text{c} \downarrow \frac{A \vee A}{A}$$

$$\text{cocontraction} \quad \text{c} \uparrow \frac{A}{A \wedge A}$$

$$\text{weakening} \quad \text{w} \downarrow \frac{\perp}{A}$$

$$\text{coweakening} \quad \text{w} \uparrow \frac{A}{\top}$$

$$\text{c} \downarrow \frac{\left(\frac{A}{A \vee \text{w} \downarrow \frac{\perp}{C}} \wedge \frac{B}{B \vee \text{w} \downarrow \frac{\perp}{D}} \right) \vee \left(\frac{C}{\text{w} \downarrow \frac{\perp}{A} \vee C} \wedge \frac{D}{\text{w} \downarrow \frac{\perp}{B} \vee D} \right)}{(A \vee C) \wedge (B \vee D)}$$

Open deduction : atomicity

Inference rules of system SKSg for propositional classical logic

$$\text{identity} \\ \text{ai} \downarrow \frac{\top}{a \vee a^\perp}$$

$$\text{cut} \\ \text{ai} \uparrow \frac{a \wedge a^\perp}{\perp}$$

$$\text{switch} \\ \text{s} \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C} = \frac{A}{B}$$

$$\text{contraction} \\ \text{c} \downarrow \frac{A \vee A}{A}$$

$$\text{cocontraction} \\ \text{c} \uparrow \frac{A}{A \wedge A}$$

$$\text{weakening} \\ \text{w} \downarrow \frac{\perp}{A}$$

$$\text{coweakening} \\ \text{w} \uparrow \frac{A}{\top}$$

$$\text{m} \frac{(A \wedge B) \vee (C \wedge D)}{(A \vee C) \wedge (B \vee D)}$$

medial

Open deduction : atomicity

Inference rules of system SKSg for propositional classical logic

$$\text{identity} \quad \text{ai} \downarrow \frac{\top}{a \vee a^\perp}$$

$$\text{cut} \quad \text{ai} \uparrow \frac{a \wedge a^\perp}{\perp}$$

$$\text{switch} \quad \text{s} \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C} = \frac{A}{B}$$

$$\text{contraction} \quad \text{c} \downarrow \frac{A \vee A}{A}$$

$$\text{cocontraction} \quad \text{c} \uparrow \frac{A}{A \wedge A}$$

$$\text{weakening} \quad \text{w} \downarrow \frac{\perp}{A}$$

$$\text{coweakening} \quad \text{w} \uparrow \frac{A}{\top}$$

$$\text{medial} \quad \text{m} \frac{(A \wedge B) \vee (A \wedge B)}{\text{c} \downarrow \frac{(A \vee A)}{A} \wedge \text{c} \downarrow \frac{(B \vee B)}{B}}$$

medial allows
to decompose
contraction to
atomic form

Open deduction : atomicity

Inference rules of system SKSg for propositional classical logic

identity

$$\text{ai}\downarrow \frac{\top}{a \vee a^\perp}$$

cut

$$\text{ai}\uparrow \frac{a \wedge a^\perp}{\perp}$$

switch

$$\text{s} \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C} = \frac{A}{B}$$

contraction

$$\text{ac}\downarrow \frac{a \vee a}{a}$$

cocontraction

$$\text{ac}\uparrow \frac{a}{a \wedge a}$$

medial

$$\text{m} \frac{(A \wedge B) \vee (C \wedge D)}{(A \vee C) \wedge (B \vee D)}$$

$$\text{uc}\downarrow \frac{\top \vee \top}{\top}$$

$$\text{uc}\uparrow \frac{\perp}{\perp \wedge \perp}$$

weakening

$$\text{w}\downarrow \frac{\perp}{A}$$

coweakening

$$\text{w}\uparrow \frac{A}{\top}$$

medial allows
to decompose
contraction to
atomic form

Open deduction : atomicity

Inference rules of system SKS for propositional classical logic

$$\text{identity} \quad \text{ai} \downarrow \frac{\top}{a \vee a^\perp}$$

$$\text{cut} \quad \text{ai} \uparrow \frac{a \wedge a^\perp}{\perp}$$

$$\text{switch} \quad \text{s} \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C} = \frac{A}{B}$$

$$\text{contraction} \quad \text{ac} \downarrow \frac{a \vee a}{a}$$

$$\text{cocontraction} \quad \text{ac} \uparrow \frac{a}{a \wedge a}$$

$$\text{medial} \quad \text{m} \frac{(A \wedge B) \vee (C \wedge D)}{(A \vee C) \wedge (B \vee D)}$$

$$\text{uc} \downarrow \frac{\top \vee \top}{\top}$$

$$\text{uc} \uparrow \frac{\perp}{\perp \wedge \perp}$$

$$\text{weakening} \quad \text{w} \downarrow \frac{\perp}{a}$$

$$\text{coweakening} \quad \text{w} \uparrow \frac{a}{\top}$$



Every structural rule is **atomic** so
every inference step is local

Open deduction : bureaucracy

$$\frac{\text{weakR} \frac{\text{contrR} \frac{\vdash A, A, \Gamma}{\vdash A, \Gamma}}{\vdash A, B, \Gamma}}{=} \frac{\text{weakR} \frac{\vdash A, A, \Gamma}{\vdash A, A, B, \Gamma}}{\text{contrR} \frac{\vdash A, B, \Gamma}}$$

$$\begin{array}{c} \swarrow \quad \searrow \\ A \vee A \vee C \\ = \frac{}{} \\ \text{c}\downarrow \frac{A \vee A}{A} \vee \text{w}\downarrow \frac{\perp}{B} \vee C \end{array}$$



In open deduction we can reduce the bureaucracy of parallel independent sub-derivations (similarly to proof nets)

Open deduction : bureaucracy

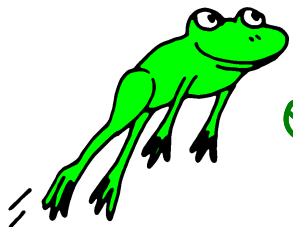
$$\text{cut} \frac{\vdash A, \Gamma \quad \text{contr} \frac{\vdash A^\perp, B, B, \Delta}{\vdash A^\perp, B, \Delta}}{\vdash \Gamma, B, \Delta}$$



$$\text{2s} \frac{(A \vee C) \wedge \left(A^\perp \vee \text{cl} \frac{B \vee B}{B} \vee D \right)}{\text{it} \frac{A \wedge A^\perp}{\perp} \vee C \vee B \vee D}$$



We can develop
explicit substitutions
for derivations



$$\text{cut} \frac{\vdash A, \Gamma \quad \vdash A^\perp, B, B, \Delta}{\text{contr} \frac{\vdash \Gamma, B, B, \Delta}{\vdash \Gamma, B, \Delta}}$$



$$\text{2s} \frac{(A \vee C) \wedge (A^\perp \vee B \vee B \vee D)}{\text{it} \frac{A \wedge A^\perp}{\perp} \vee C \vee \text{cl} \frac{B \vee B}{B} \vee D}$$



$$\left\langle \text{cl} \frac{B \vee B}{B} / X \right\rangle \text{2s} \frac{(A \vee C) \wedge (A^\perp \vee X \vee D)}{\text{it} \frac{A \wedge A^\perp}{\perp} \vee C \vee X \vee D}$$

Open deduction

Derivations are symmetric

Inference rules are atomic or linear

Derivations are low in bureaucracy

We can see the geometric nature of derivations clearly
and develop normalisation procedures.

From derivations to flows

$$ai\downarrow \frac{T}{a \vee a^\perp} \quad \text{diagram: a horizontal line with a vertical line above it, labeled } + \text{ on the left and } - \text{ on the right}$$

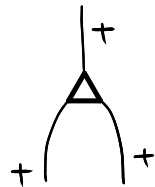


$$ai\uparrow \frac{a \wedge a^\perp}{\perp}$$

$$s \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C}$$

$$uc\downarrow \frac{T \vee T}{T}$$

$$ac\downarrow \frac{a \vee a^\perp}{a} \quad \text{diagram: a vertical line with a triangle pointing down, labeled + on the left and + on the right}$$



$$ac\uparrow \frac{a}{a \wedge a}$$

$$m \frac{(A \wedge B) \vee (C \wedge D)}{(A \vee C) \wedge (B \vee D)}$$

$$uc\uparrow \frac{\perp}{\perp \wedge \perp}$$

$$w\downarrow \frac{\perp}{a} \quad \text{diagram: a vertical line with a triangle pointing down, labeled + on the left}$$



$$w\uparrow \frac{a}{T}$$

$$= \frac{A}{B}$$



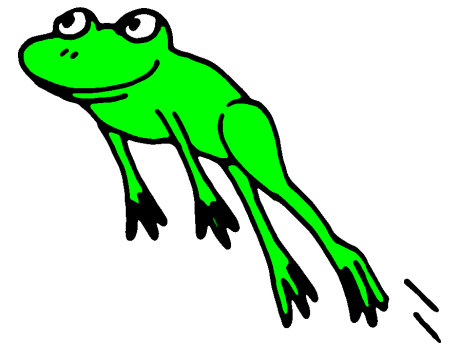
We associate a generator to each structural rule

Summary

We introduced open deduction and saw how **symmetry**, **atomicity**, and **low bureaucracy** lead to **good normalisation** (for propositional classical logic)

Tomorrow : Normalisation for linear logics

Proof compression mechanisms



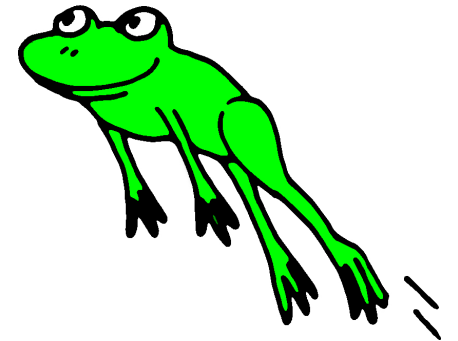
References

Linear Logic and Noncommutativity in the Calculus of Structures, Lutz Straßburger's PhD thesis, 2003

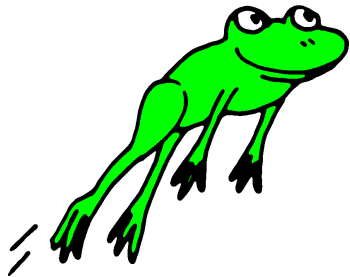
A system of Interaction and Structure, ACM Transactions on Computational Logic 8 (1:1) 2007, pp. 1–64,
Alessio Guglielmi

Deep Inference and Symmetry in Classical Proofs, Kai Brünnler's PhD thesis, 2004

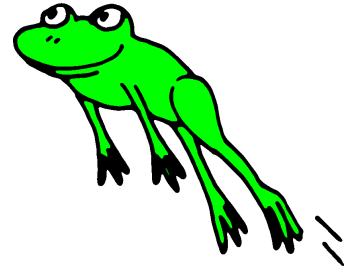
A General View of Normalisation Through Atomic Flows, Tom Gundersen's PhD thesis, 2010



An introduction to deep inference



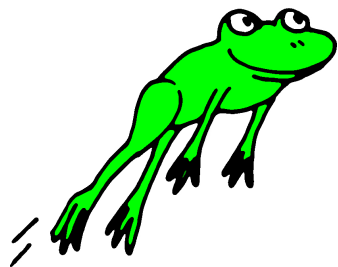
Victoria Barrett
Inria Saclay, Équipe Partout



Proof Society 2026, Aussois

Today : Normalisation via **atomic flows** for
propositional classical logic

Normalisation via **splitting** and
decomposition



Atomic Flows

From derivations to flows

$$ai\downarrow \frac{T}{a \vee a^\perp} \quad \text{diagram: a horizontal line with a vertical line above it, labeled } + \text{ on the left and } - \text{ on the right}$$

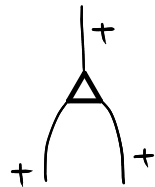


$$ai\uparrow \frac{a \wedge a^\perp}{\perp}$$

$$s \frac{A \wedge (B \vee C)}{(A \wedge B) \vee C} \quad ||| ||| |||$$

$$uc\downarrow \frac{T \vee T}{T}$$

$$ac\downarrow \frac{a \vee a^\perp}{a} \quad \text{diagram: a Y-shaped junction with two inputs at the top labeled } + \text{ and } + \text{, and one output at the bottom labeled } +$$



$$ac\uparrow \frac{a}{a \wedge a}$$

$$m \frac{(A \wedge B) \vee (C \wedge D)}{(A \vee C) \wedge (B \vee D)}$$

$$uc\uparrow \frac{\perp}{\perp \wedge \perp}$$

$$w\downarrow \frac{\perp}{a} \quad \text{diagram: a vertical line with a triangle at the bottom, labeled } + \text{ below the triangle}$$



$$w\uparrow \frac{a}{T}$$

$$||| \text{ (crossed) } ||| = ||| ||| |||$$

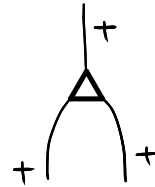
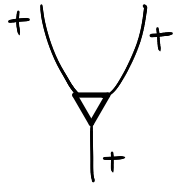
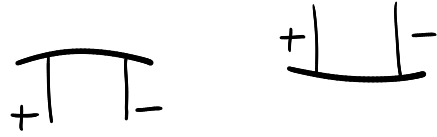
$$= \frac{A}{B} \quad |||$$



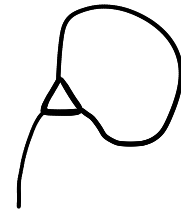
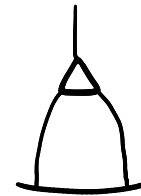
We associate a generator to each structural rule

From derivations to flows

Atomic flows are graphs whose nodes are labelled by generators and whose edges are polarised and vertically directed



Forbidden :



We associate a generator to each structural rule

From derivations to flows

$$\begin{array}{c}
 \text{ac} \uparrow \frac{a}{a \wedge a} \quad \vee \quad \text{ac} \uparrow \frac{a^\perp}{a^\perp \wedge a^\perp} \\
 \hline
 \text{m} \frac{(a \vee a^\perp) \wedge (a \vee a^\perp)}{s} \\
 \hline
 s \frac{a^\perp \wedge (a \vee a^\perp)}{a \vee s} \\
 \hline
 \text{ai} \uparrow \frac{a^\perp \wedge a}{\perp} \quad \vee \quad \text{aw} \uparrow \frac{a^\perp}{\top}
 \end{array}$$

From derivations to flows

$$\begin{array}{c}
 \begin{array}{c}
 \text{ac} \uparrow \frac{a}{a \wedge a} \\
 \vee \quad \text{ac} \uparrow \frac{a^\perp}{a^\perp \wedge a^\perp}
 \end{array} \\
 \hline
 m \\
 (a \vee a^\perp) \wedge (a \vee a^\perp) \\
 \hline
 s \\
 a \vee s \frac{a^\perp \wedge (a \vee a^\perp)}{a^\perp \wedge a} \vee \text{aw} \uparrow \frac{a^\perp}{\perp}
 \end{array}$$

From derivations to flows

$$\begin{array}{c}
 \begin{array}{c}
 \text{ac} \uparrow \\
 \hline
 \begin{array}{c}
 a \\
 \wedge \\
 a
 \end{array}
 \end{array}
 \quad \vee \quad
 \begin{array}{c}
 \text{ac} \uparrow \\
 \hline
 \begin{array}{c}
 a^\perp \\
 \wedge \\
 a^\perp
 \end{array}
 \end{array}
 \\
 \hline
 m \\
 (a \vee a^\perp) \wedge (a \vee a^\perp) \\
 \hline
 s \\
 a \vee s \frac{a^\perp \wedge (a \vee a^\perp)}{a^\perp \wedge a} \quad \vee \quad \text{aw} \uparrow \frac{a^\perp}{\top}
 \end{array}$$

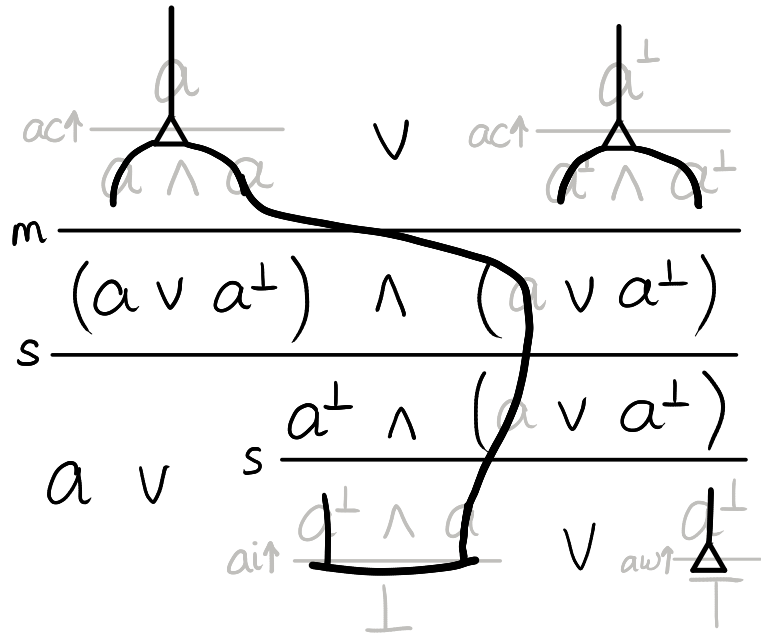
From derivations to flows

$$\begin{array}{c}
 \begin{array}{c}
 \text{ac} \uparrow \\
 \hline
 \begin{array}{c}
 a \\
 \wedge \\
 a
 \end{array}
 \end{array}
 \quad \vee \quad
 \begin{array}{c}
 \text{ac} \uparrow \\
 \hline
 \begin{array}{c}
 a^\perp \\
 \wedge \\
 a^\perp
 \end{array}
 \end{array}
 \\
 \hline
 m \\
 (a \vee a^\perp) \wedge (a \vee a^\perp) \\
 \hline
 s \\
 a \vee s \frac{a^\perp \wedge (a \vee a^\perp)}{a^\perp \wedge a} \vee \frac{a^\perp}{\perp}
 \end{array}$$

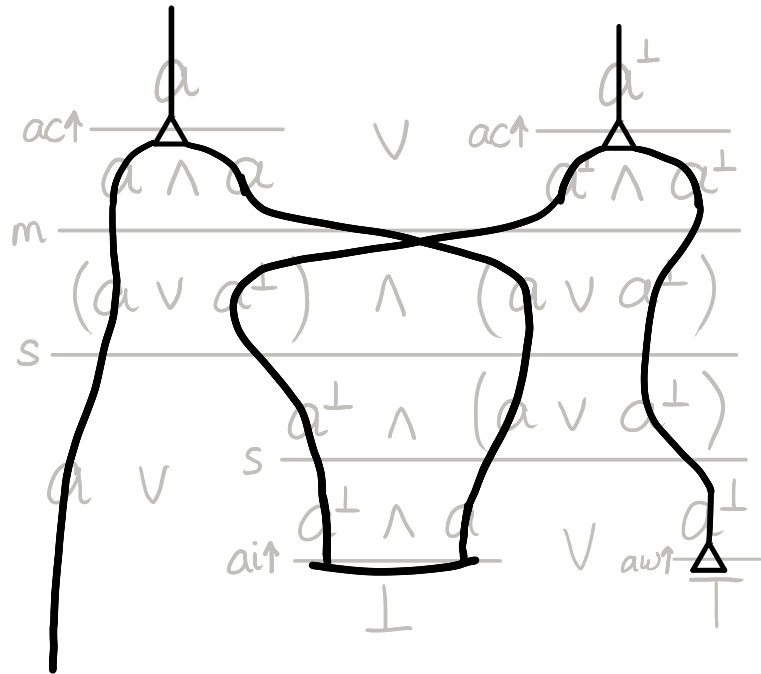
From derivations to flows

$$\begin{array}{c}
 \begin{array}{c}
 \text{ac} \uparrow \\
 \text{---} \\
 \begin{array}{c} \diagup \quad \diagdown \\ a \quad a \end{array} \\
 \text{---} \\
 a \wedge a
 \end{array}
 \quad \vee \quad
 \begin{array}{c}
 \text{ac} \uparrow \\
 \text{---} \\
 \begin{array}{c} \diagup \quad \diagdown \\ a^\perp \quad a^\perp \end{array} \\
 \text{---} \\
 a^\perp \wedge a^\perp
 \end{array}
 \\
 \hline
 m \\
 (a \vee a^\perp) \wedge (a \vee a^\perp) \\
 \hline
 s \\
 a \vee s \frac{a^\perp \wedge (a \vee a^\perp)}{a^\perp \wedge a} \vee \frac{a^\perp \wedge a^\perp}{a^\perp}
 \end{array}$$

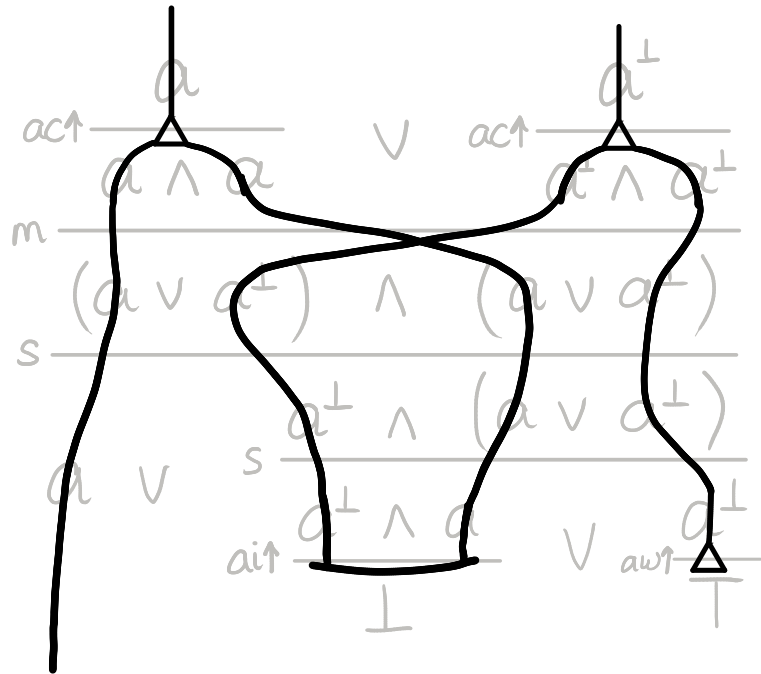
From derivations to flows



From derivations to flows

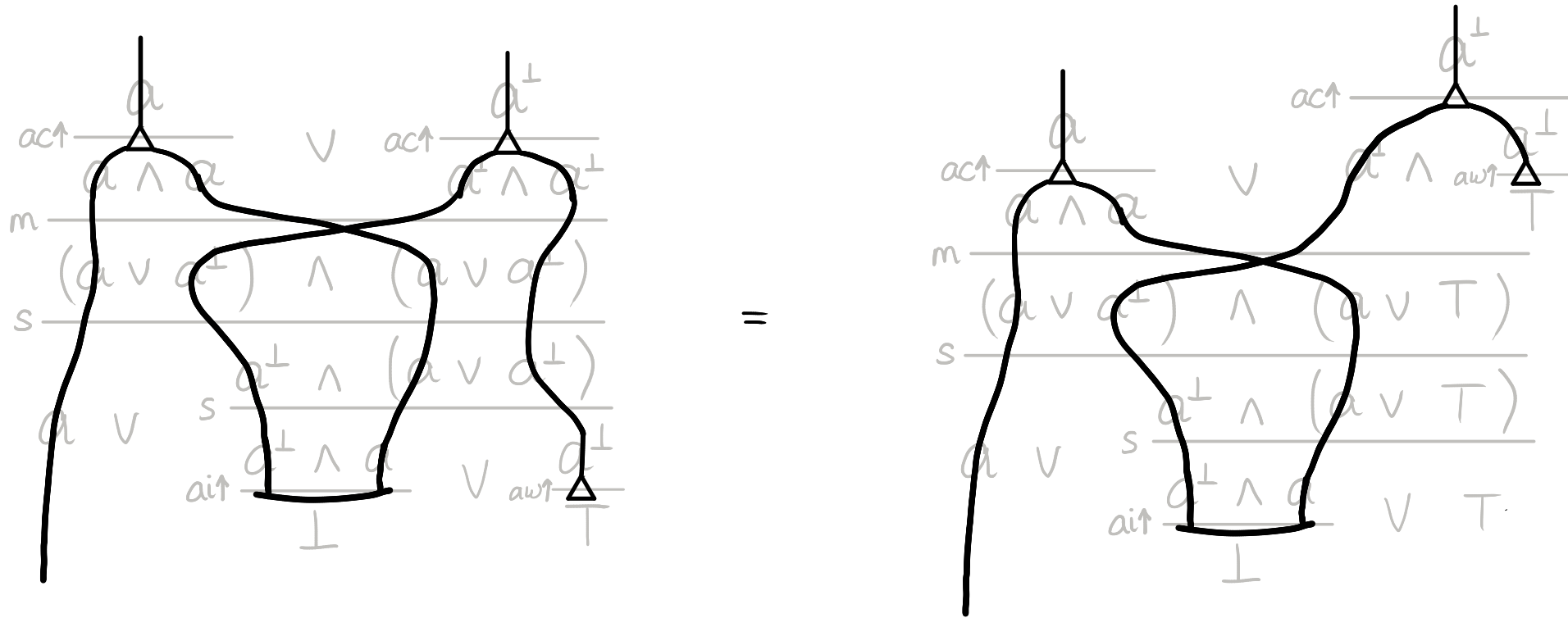


From derivations to flows



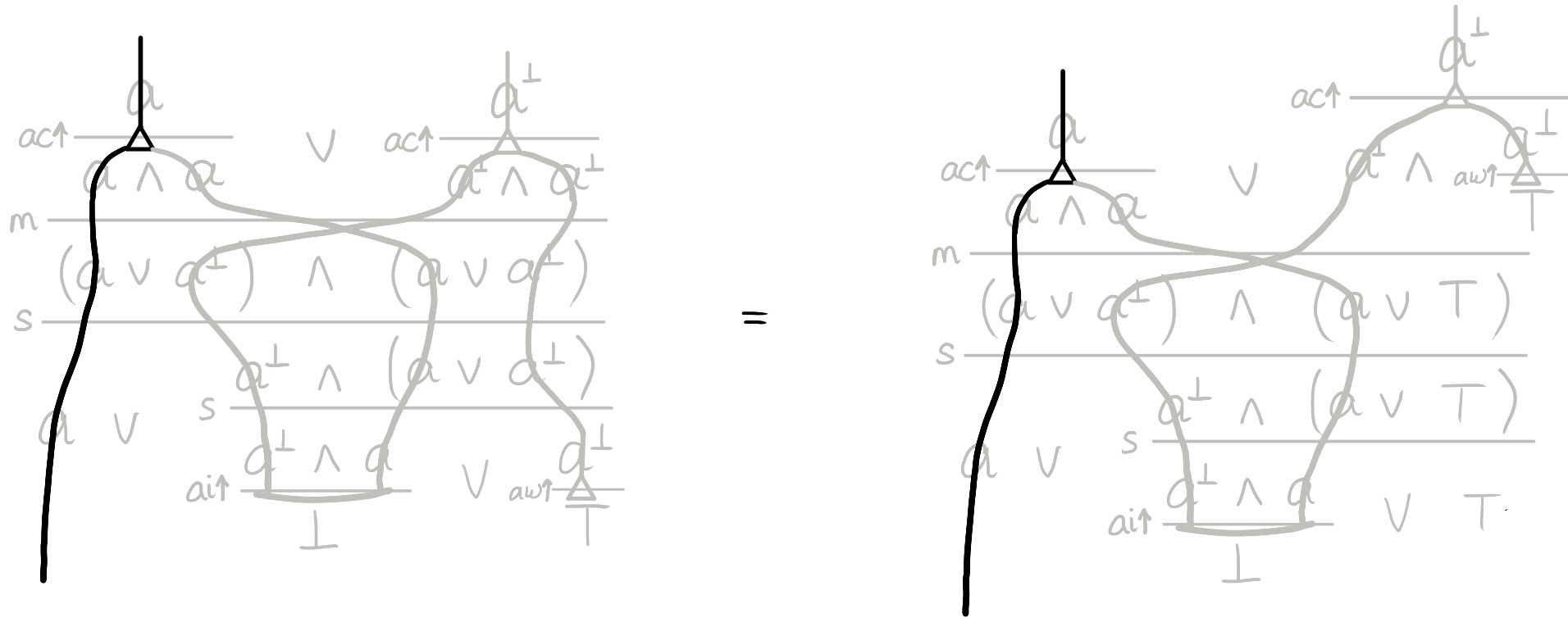
$$\begin{array}{c}
 \text{ac} \uparrow \frac{a}{a \wedge a} \quad \vee \quad \text{ac} \uparrow \frac{a^\perp}{a^\perp \wedge \text{aw} \uparrow \frac{a^\perp}{\top}} \\
 \hline
 \text{m} \frac{(a \vee a^\perp) \wedge (a \vee \top)}{s} \\
 \hline
 a \vee s \frac{a^\perp \wedge (a \vee \top)}{\text{ai} \uparrow \frac{a^\perp \wedge a}{\perp} \vee \top}
 \end{array}$$

From derivations to flows



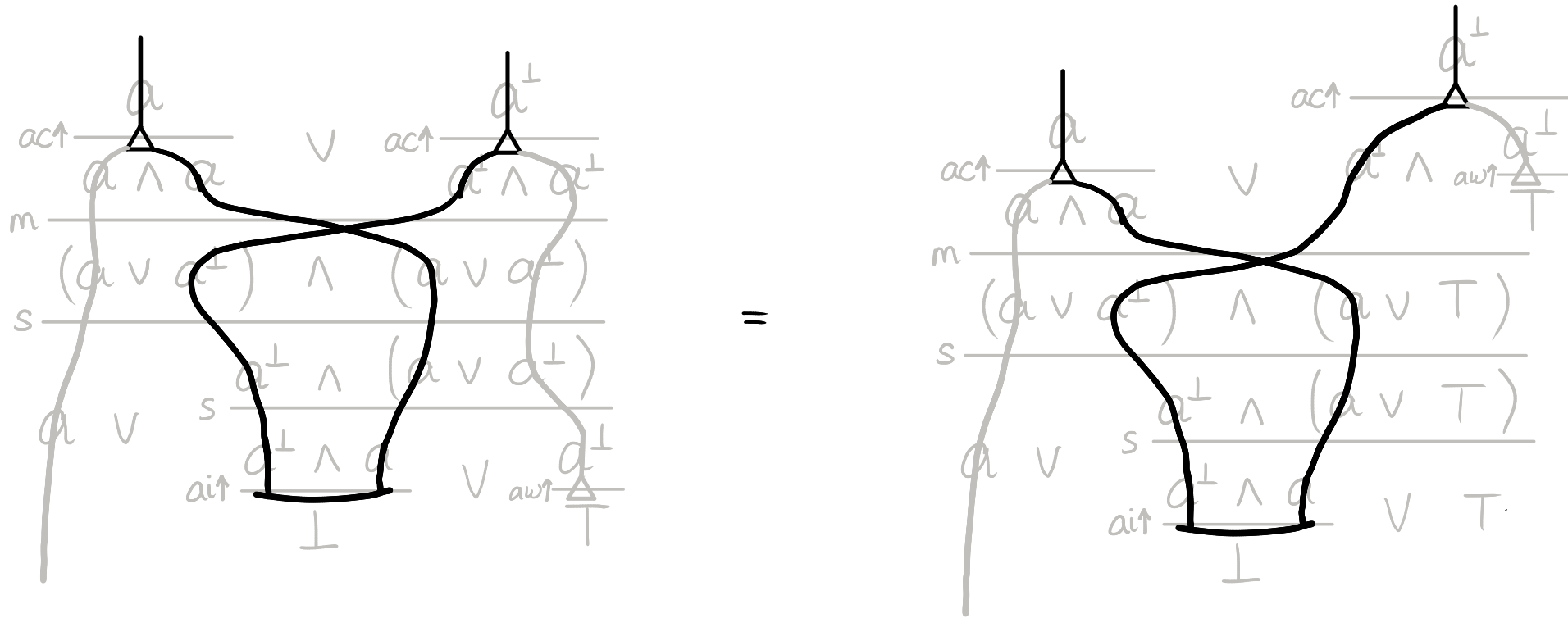
We retain only the structural information of derivations, up to continuous deformation of the flows

From derivations to flows



We retain only the structural information of derivations, up to continuous deformation of the flows

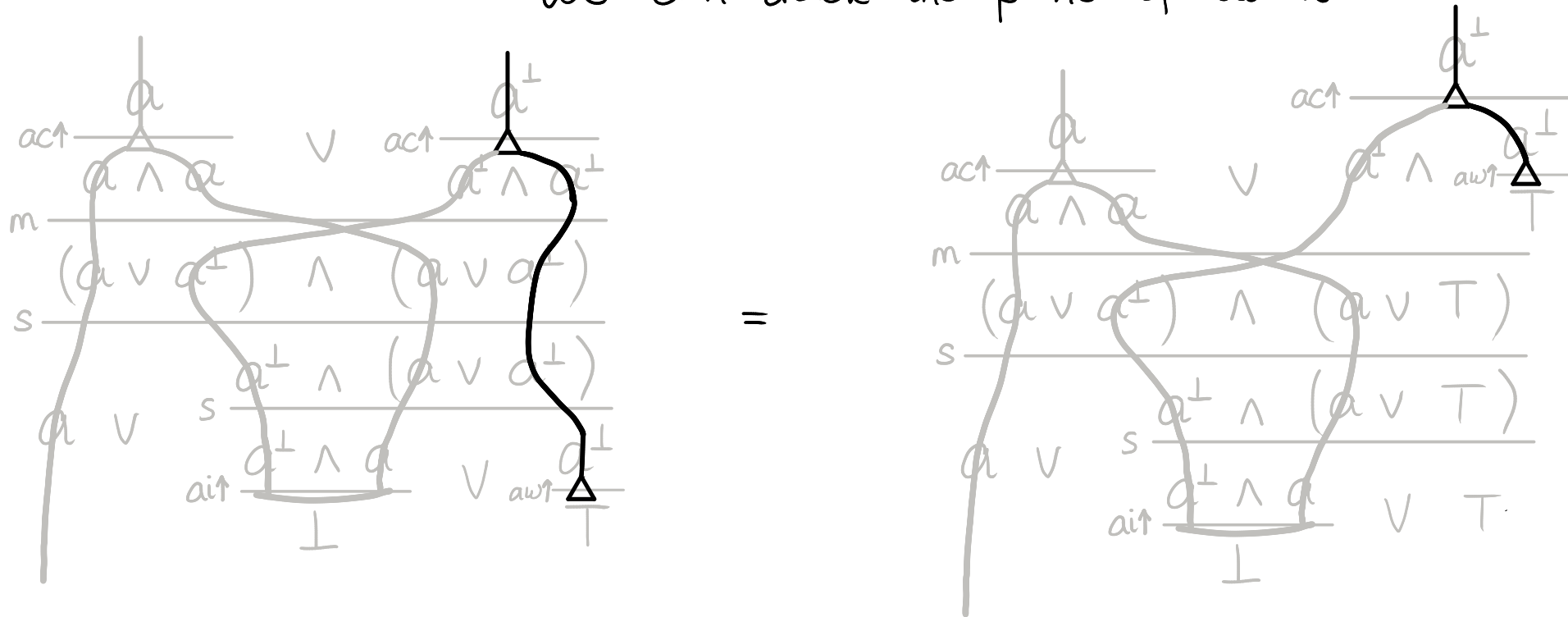
From derivations to flows



We retain only the structural information of derivations, up to continuous deformation of the flows

From derivations to flows

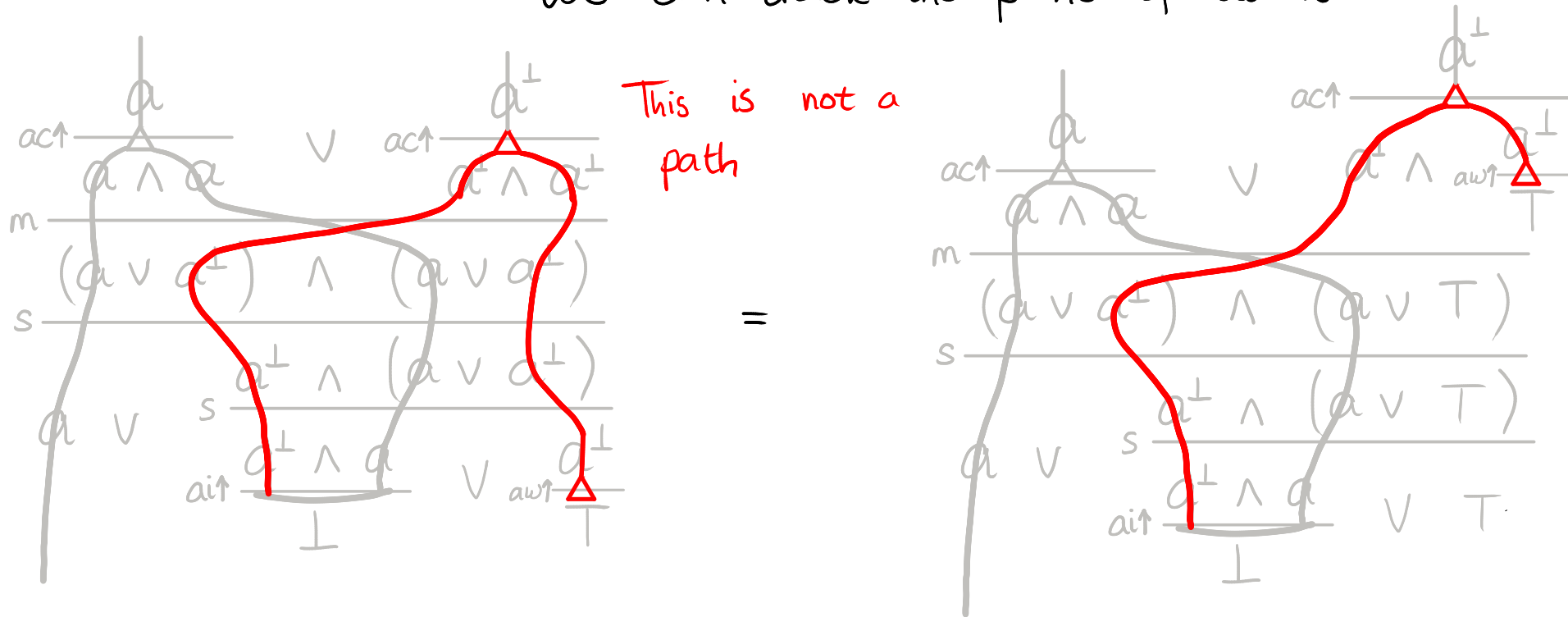
We can track the paths of atoms



We retain only the structural information of derivations, up to continuous deformation of the flows

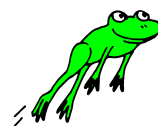
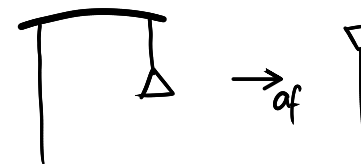
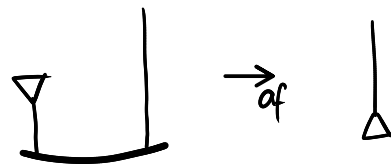
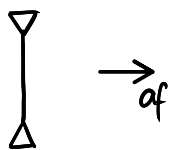
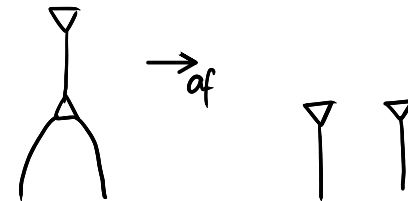
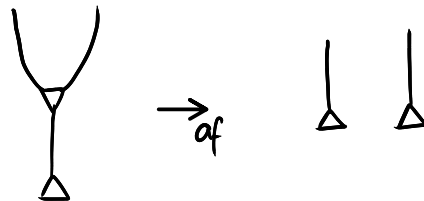
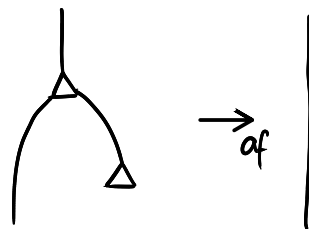
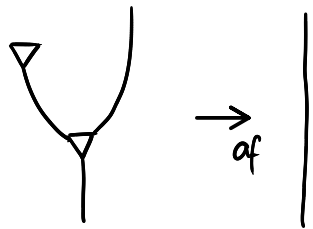
From derivations to flows

We can track the paths of atoms

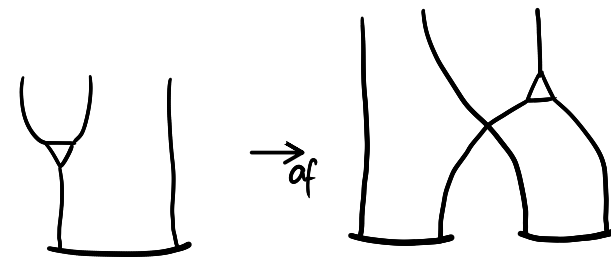
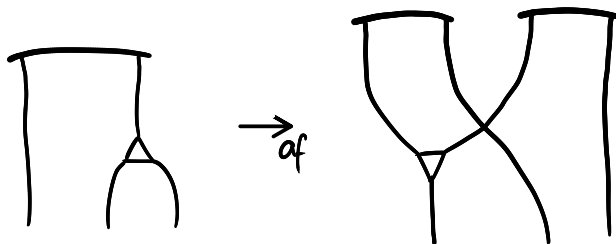
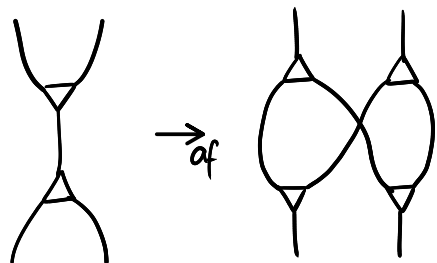


We retain only the structural information of derivations, up to continuous deformation of the flows

Rewriting flows



These ones are possible because of the **symmetry** of the rules



Rewriting flows lifts to derivations

$$K \left\{ \text{ac} \downarrow \frac{a \vee a}{a} \right\}$$

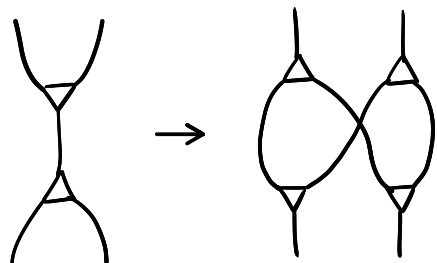
$$\Phi(a) \parallel$$

$$H \left\{ \text{ac} \uparrow \frac{a}{a \wedge a} \right\}$$

$$K \left\{ \begin{array}{c} \text{ac} \uparrow \frac{a}{a \wedge a} \vee \text{ac} \uparrow \frac{a}{a \wedge a} \\ m \frac{\quad}{\quad} \\ \text{ac} \downarrow \frac{a \vee a}{a} \wedge \text{ac} \downarrow \frac{a \vee a}{a} \end{array} \right\}$$

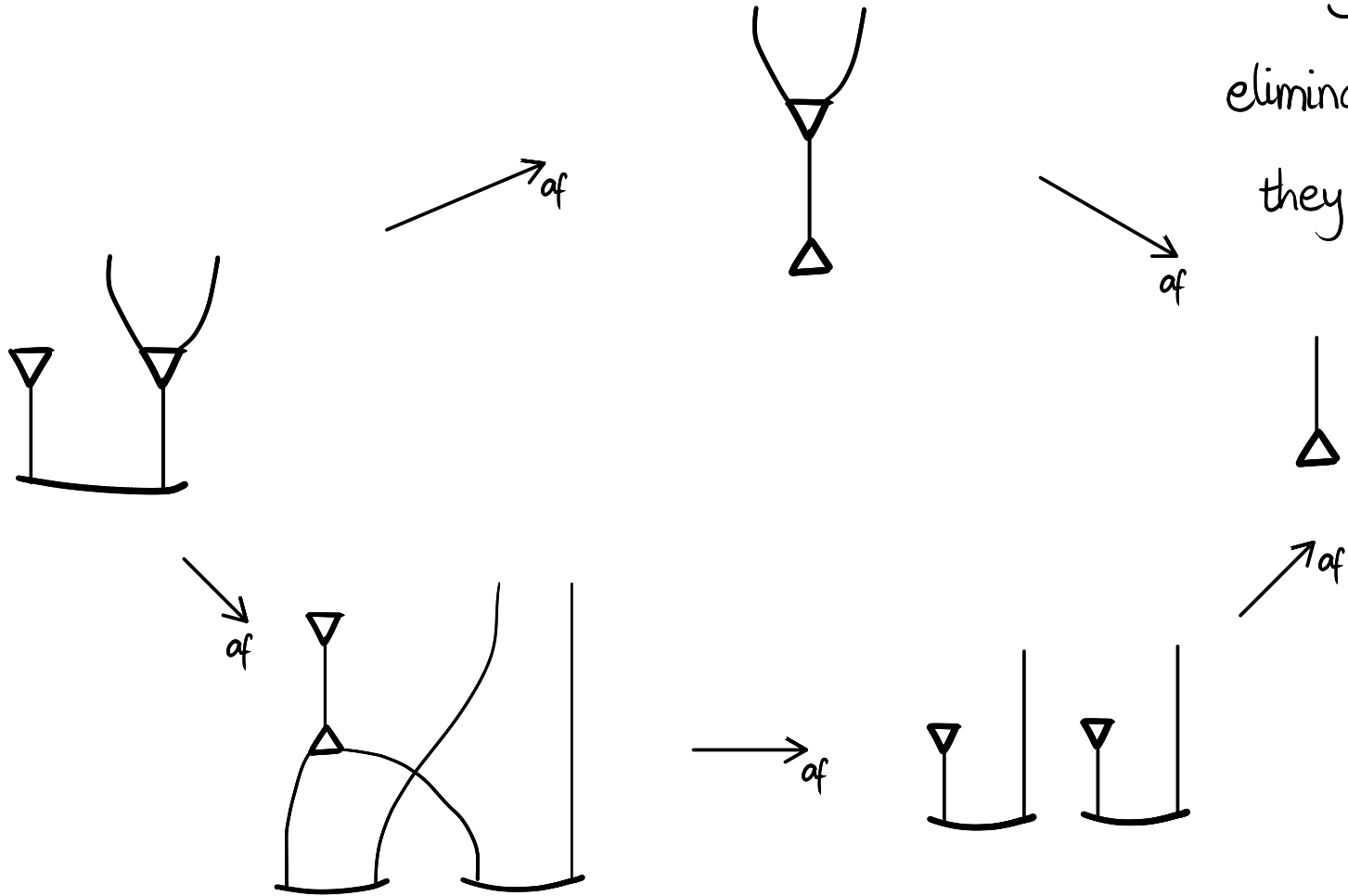
$$\Phi(a \wedge a) \parallel$$

$$H\{a \wedge a\}$$



Rewriting flows is locally confluent

Sometimes rewriting steps eliminate cuts and sometimes they duplicate them



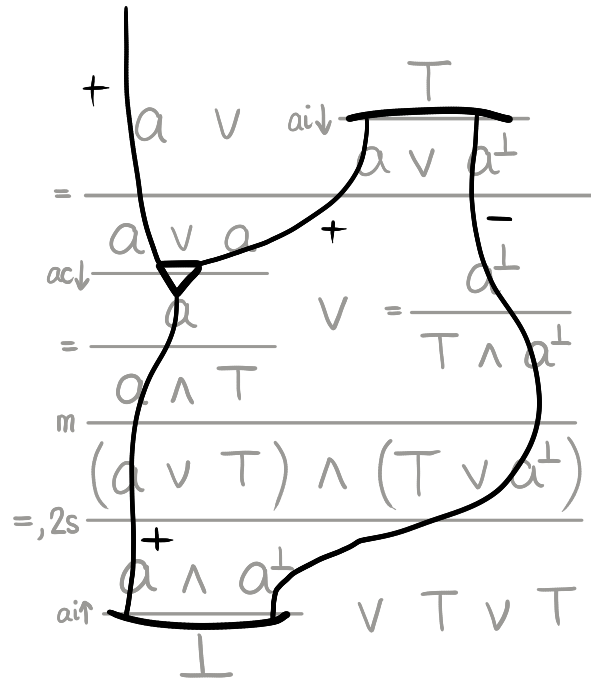
Rewriting flows doesn't always terminate

$$\begin{aligned}
 & a \vee \text{ai} \downarrow \frac{T}{a \vee a^\perp} \\
 = & \frac{}{} \\
 \text{ac} \downarrow & \frac{a \vee a}{a} \vee = \frac{a^\perp}{T \wedge a^\perp} \\
 = & \frac{a \wedge T}{a \wedge T} \\
 \text{m} & \frac{}{} \\
 =, 2s & \frac{(a \vee T) \wedge (T \vee a^\perp)}{} \\
 \text{ai} \uparrow & \frac{a \wedge a^\perp}{\perp} \vee T \vee T
 \end{aligned}$$

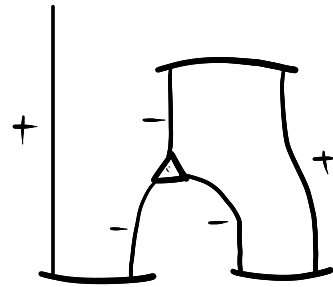
Rewriting flows doesn't always terminate

$$\begin{array}{l}
 \begin{array}{c} + \\ a \vee a_i \downarrow \end{array} \quad \begin{array}{c} T \\ a \vee a^\perp \end{array} \\
 = \frac{\begin{array}{c} a \vee a \\ a \end{array} \quad \begin{array}{c} + \\ a^\perp \end{array}}{\begin{array}{c} a \vee T \\ a \wedge T \end{array}} \quad \begin{array}{c} - \\ a^\perp \end{array} \\
 = \frac{\begin{array}{c} a \vee T \\ a \wedge T \end{array}}{m} \quad \begin{array}{c} V = \frac{a^\perp}{T \wedge a^\perp} \end{array} \\
 =, 2s \quad \frac{\begin{array}{c} + \\ a \wedge a^\perp \end{array}}{\begin{array}{c} a_i \uparrow \\ \perp \end{array}} \quad \begin{array}{c} (a \vee T) \wedge (T \vee a^\perp) \\ V \quad T \quad V \quad T \end{array}
 \end{array}$$

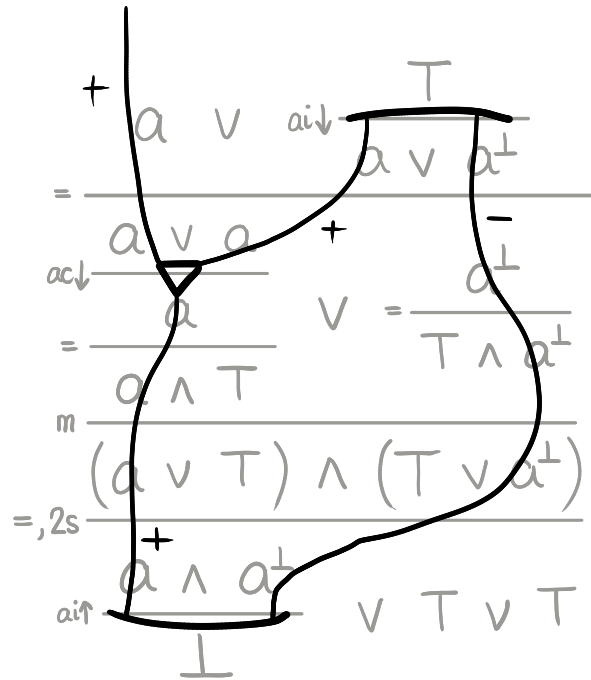
Rewriting flows doesn't always terminate



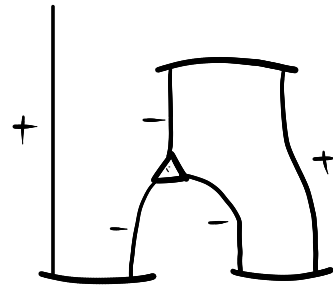
$\rightarrow_{af}$



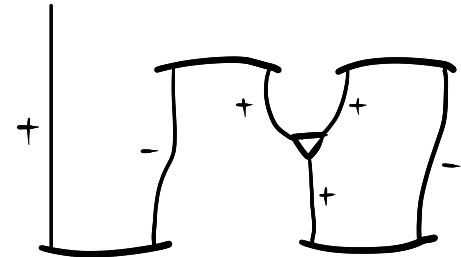
Rewriting flows doesn't always terminate



$\rightarrow_{af}$

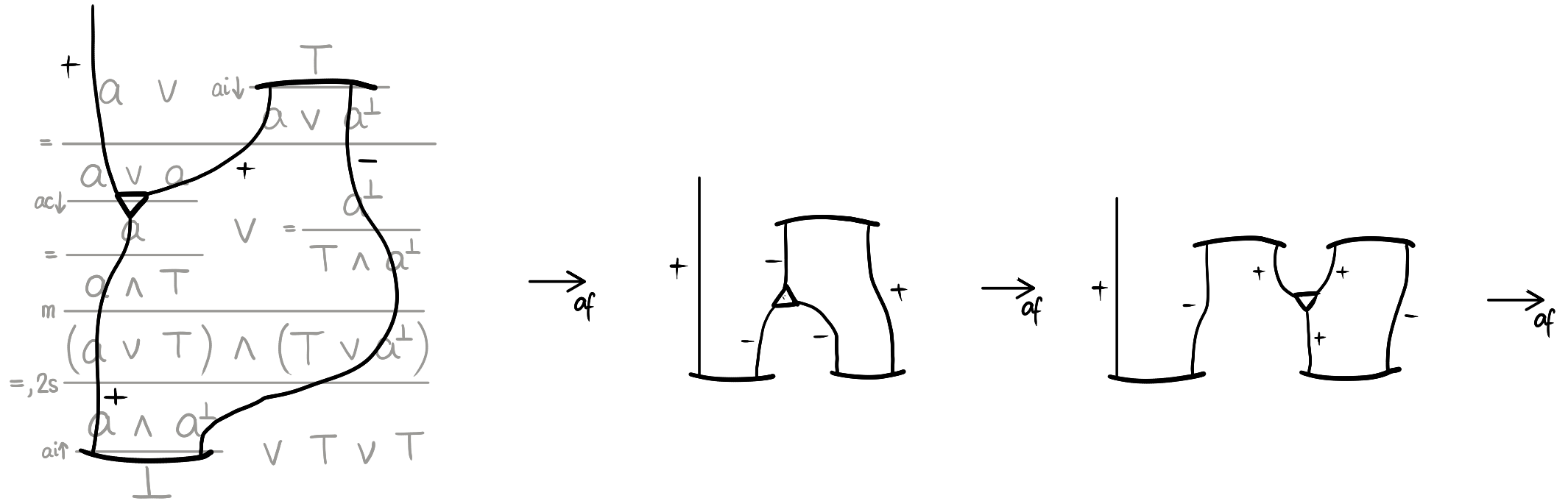


$\rightarrow_{af}$



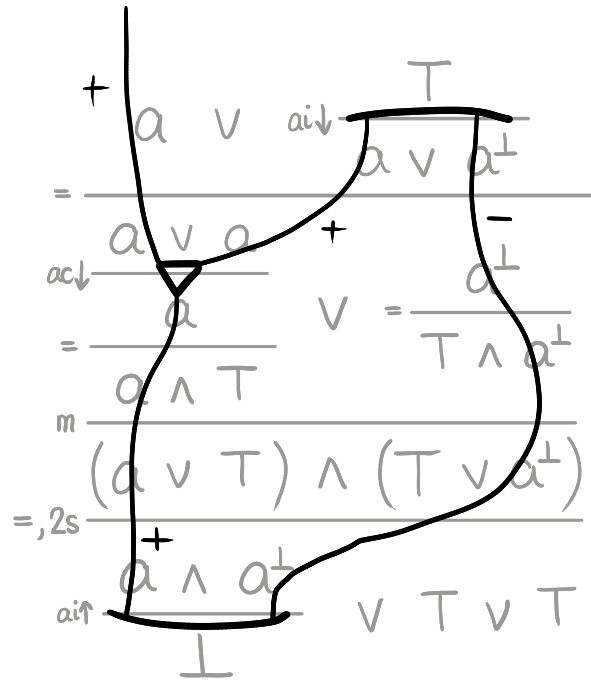
$\rightarrow_{af}$

Rewriting flows doesn't always terminate

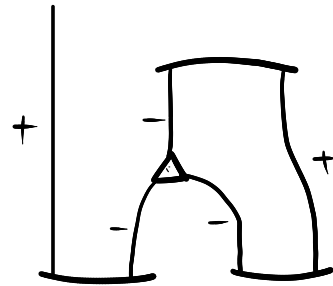


The problem arises when a (co-)contraction gets into a path between an identity and a cut

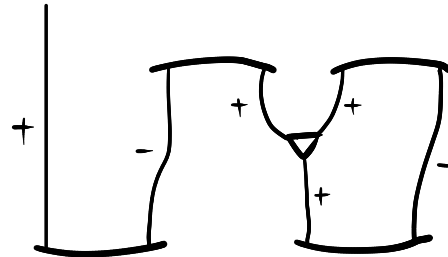
Rewriting flows doesn't always terminate



$\rightarrow_{af}$



$\rightarrow_{af}$



$\rightarrow_{af}$

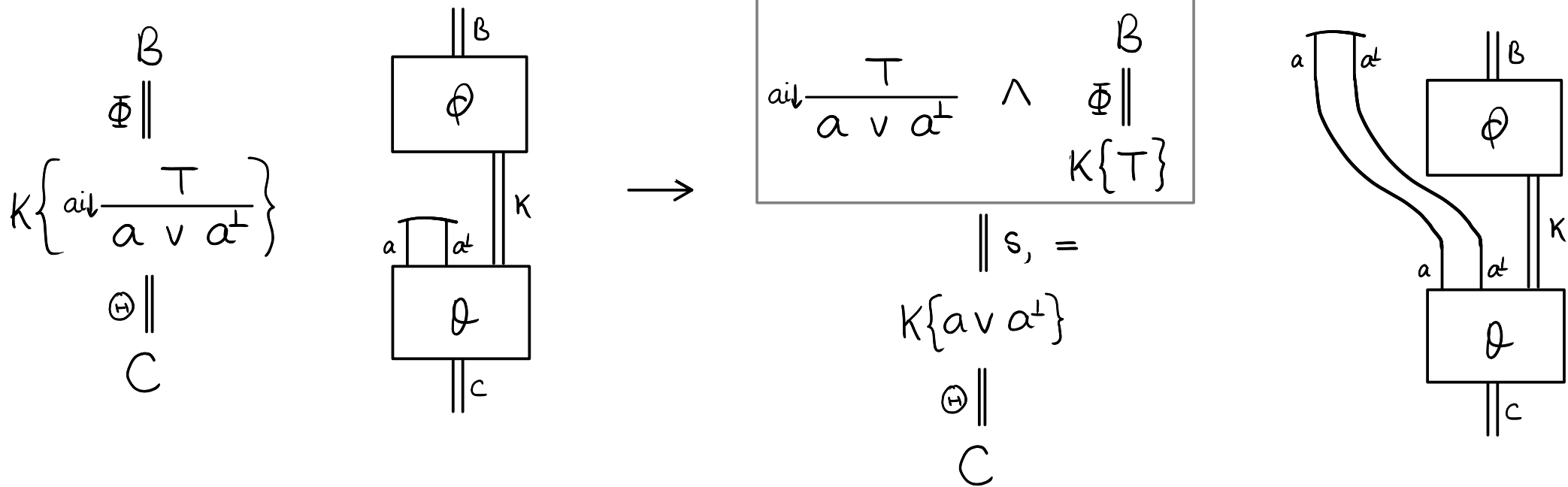
The problem arises when a (co-)contraction gets into a path between an identity and a cut



We should break the paths between identities and cuts

Breaking paths in atomic flows

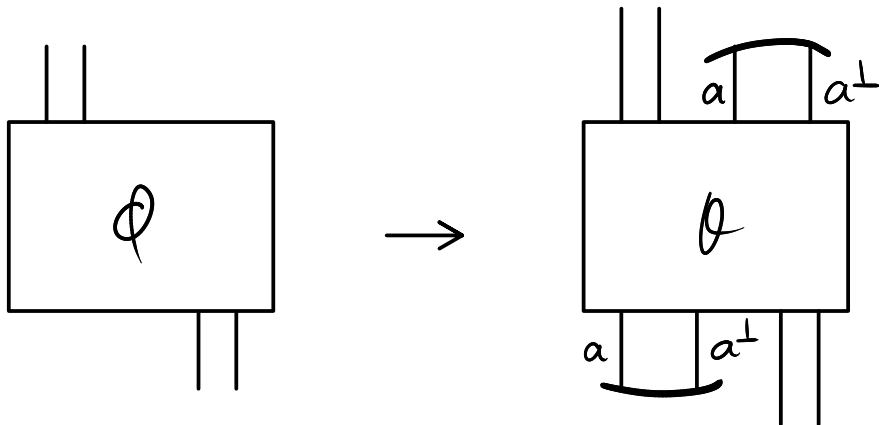
We can pull identities up derivations



Breaking paths in atomic flows

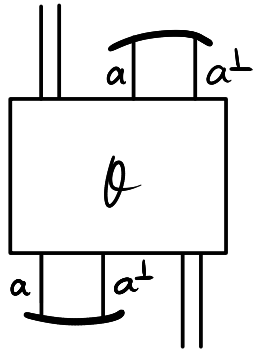
We can gather all the cuts and identities on a single atom

$$\begin{array}{c} \overline{a} \mid \overline{a^\perp} \quad \overline{a} \mid \overline{a^\perp} \\ \text{ai}\downarrow \frac{T}{a \vee a^\perp} \quad \wedge \quad \text{ai}\downarrow \frac{T}{a \vee a^\perp} \end{array} \rightarrow \begin{array}{c} \text{ai}\downarrow \frac{T}{a} \quad \vee \quad \text{ai}\downarrow \frac{T}{a^\perp} \\ \text{act} \frac{a \wedge a}{(a \vee a^\perp)} \quad \wedge \quad \text{act} \frac{a^\perp \wedge a^\perp}{(a \vee a^\perp)} \\ \text{m} \end{array} \quad \begin{array}{c} \overline{a} \mid \overline{a^\perp} \\ \triangle \quad \triangle \\ \text{cur} \end{array}$$

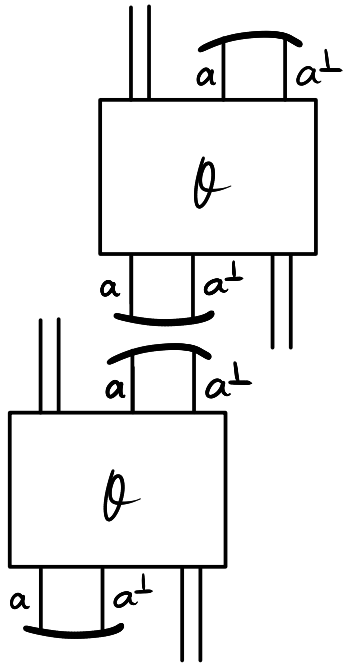
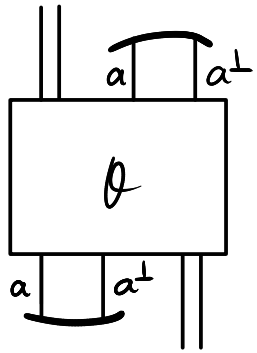


where Q contains no cuts or identities on a

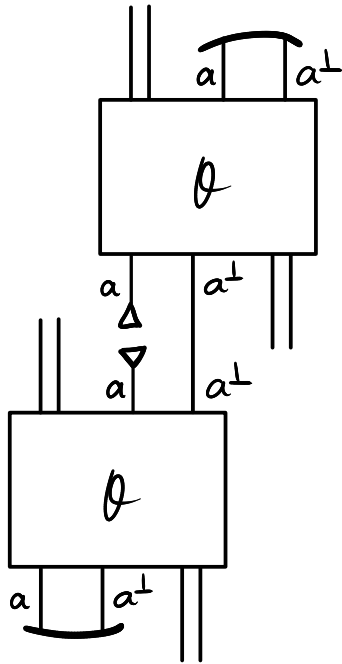
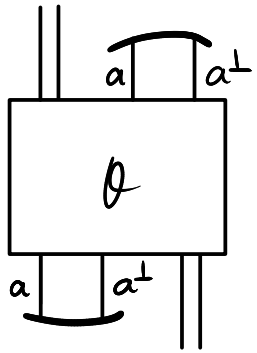
Breaking paths in atomic flows



Breaking paths in atomic flows

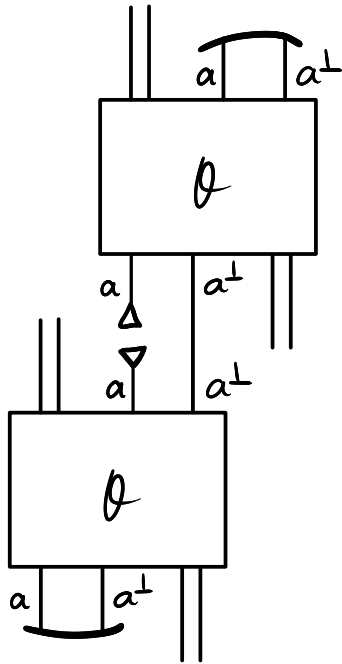
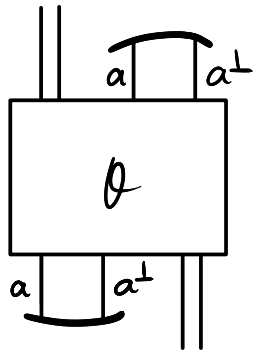


Breaking paths in atomic flows



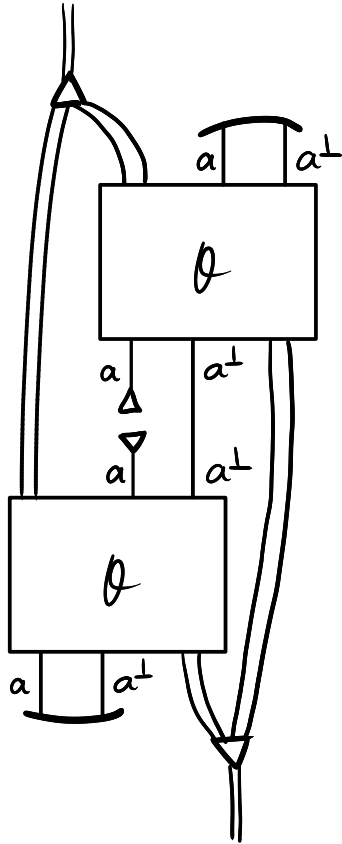
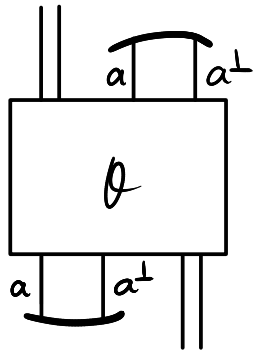
$$\begin{aligned} & a\omega \uparrow \frac{a}{T} \wedge a^\perp \\ & = \frac{1}{a} \vee a^\perp \\ & a\omega \downarrow \frac{1}{a} \vee a^\perp \end{aligned}$$

Breaking paths in atomic flows



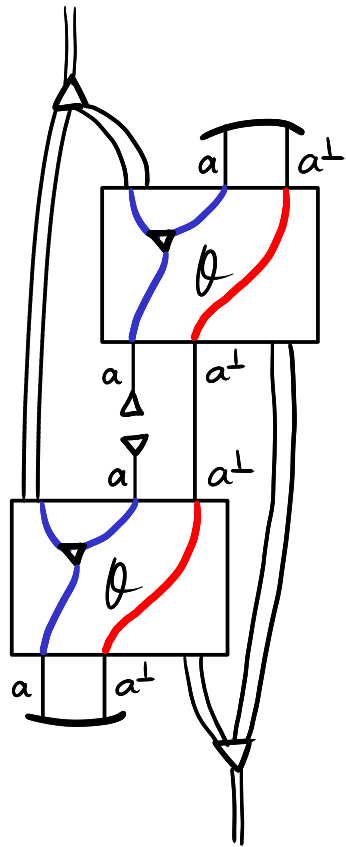
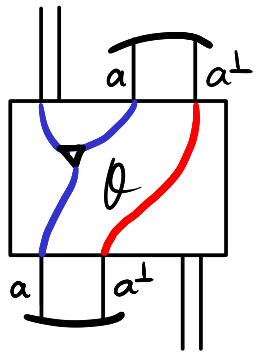
$$\begin{aligned}
 & B \wedge \text{ai} \downarrow \frac{T}{a \vee a^\perp} \\
 & \quad \oplus \parallel \\
 & B \wedge \left(\begin{aligned} & \text{aw} \uparrow \frac{a}{T} \wedge a^\perp \\ & = \frac{\perp}{\text{aw} \downarrow \frac{1}{a} \vee a^\perp} \vee C \end{aligned} \right) \\
 & \quad \text{---} \quad \frac{s}{(B \wedge (a \vee a^\perp)) \vee C} \\
 & \quad \quad \oplus \parallel \\
 & \quad \text{ai} \uparrow \frac{a \wedge a^\perp}{\perp} \vee C
 \end{aligned}$$

Breaking paths in atomic flows

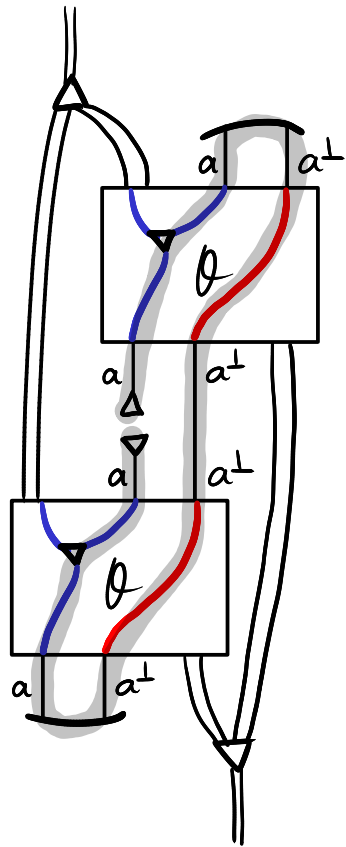
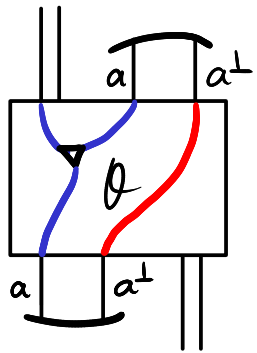


$$\begin{aligned}
 &=, c \uparrow \frac{B}{B \wedge \frac{a \downarrow \frac{T}{a \vee a^\perp}}{\ominus \parallel}} \\
 &B \wedge \left(\frac{a \uparrow \frac{a}{T} \wedge a^\perp}{a \downarrow \frac{1}{a} \vee a^\perp} \vee C \right) \\
 &\frac{s}{(B \wedge (a \vee a^\perp)) \vee C} \\
 &\ominus \parallel \\
 &a \uparrow \frac{a \wedge a^\perp}{1} \vee C \\
 &=, c \downarrow \frac{1}{C}
 \end{aligned}$$

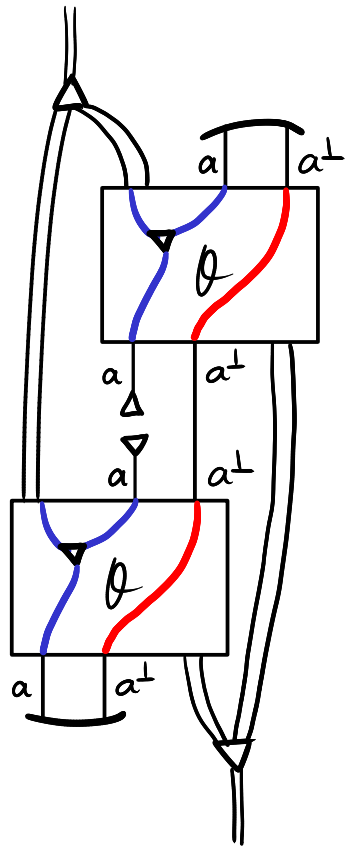
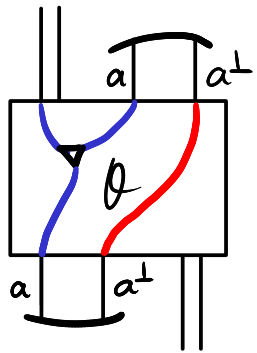
Breaking paths in atomic flows



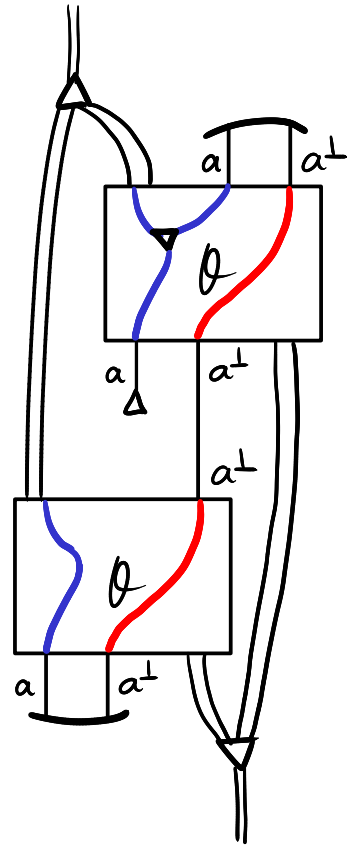
Breaking paths in atomic flows



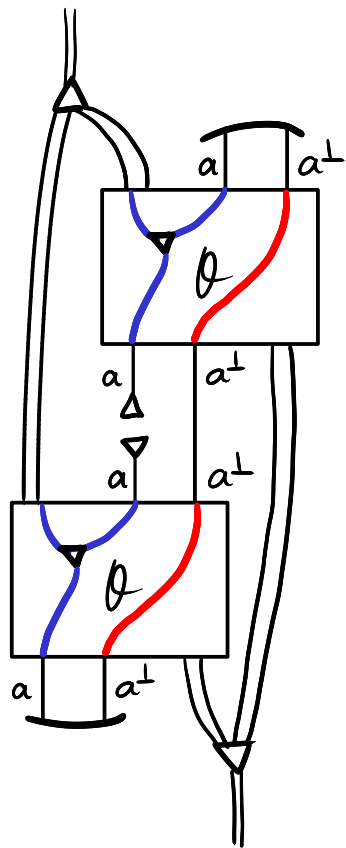
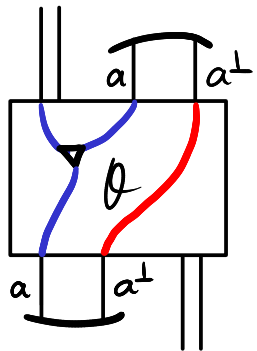
Breaking paths in atomic flows



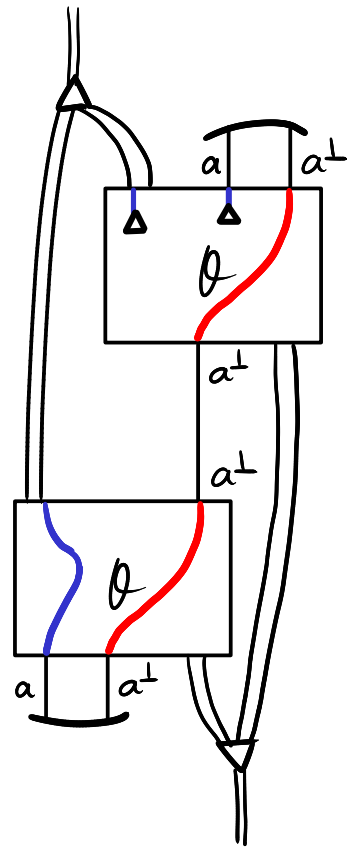
$\rightarrow_{af^*}$



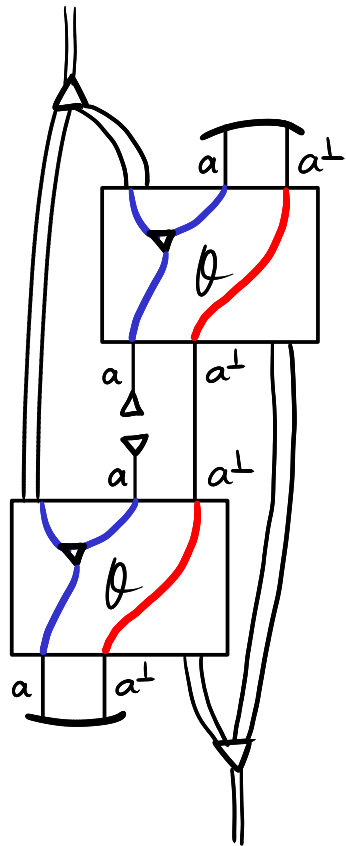
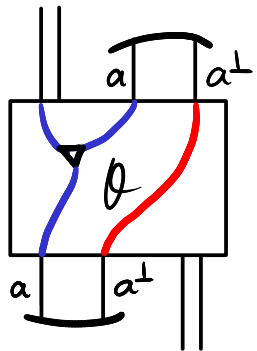
Breaking paths in atomic flows



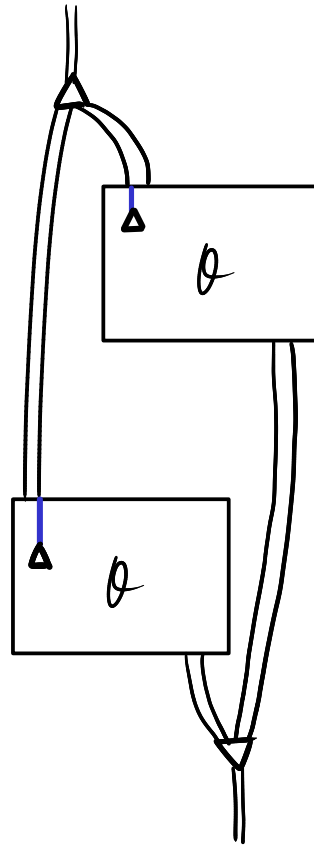
$\xrightarrow{af^*}$



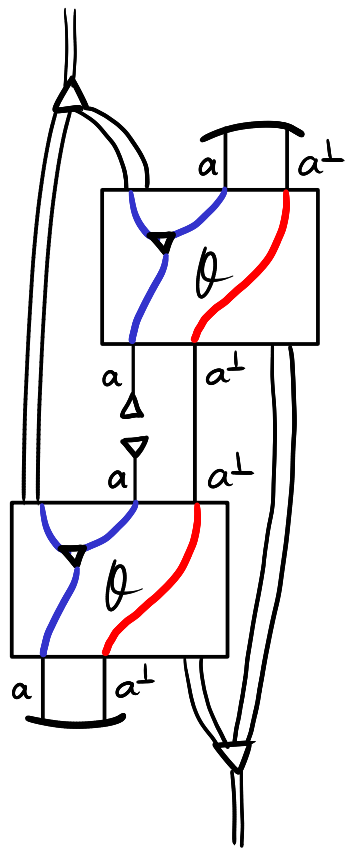
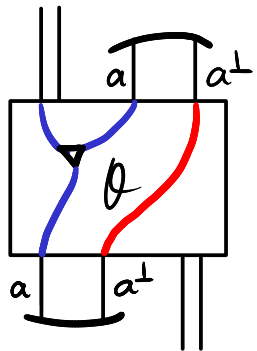
Breaking paths in atomic flows



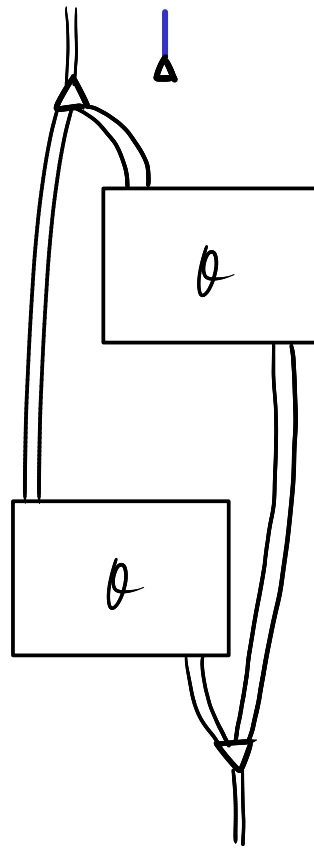
$\xrightarrow{af^*}$



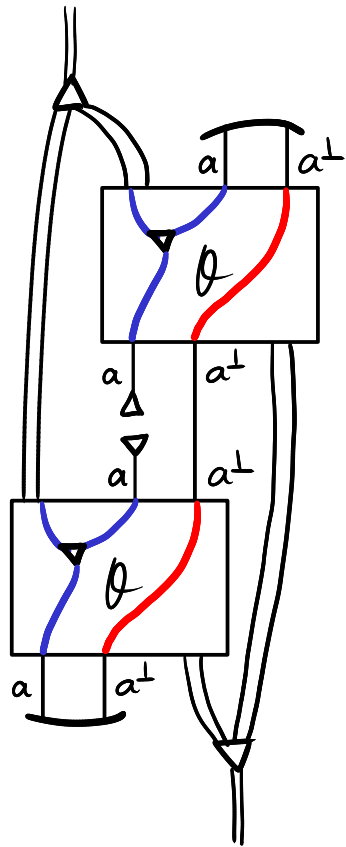
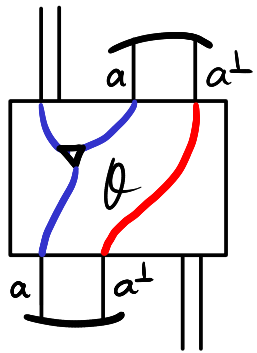
Breaking paths in atomic flows



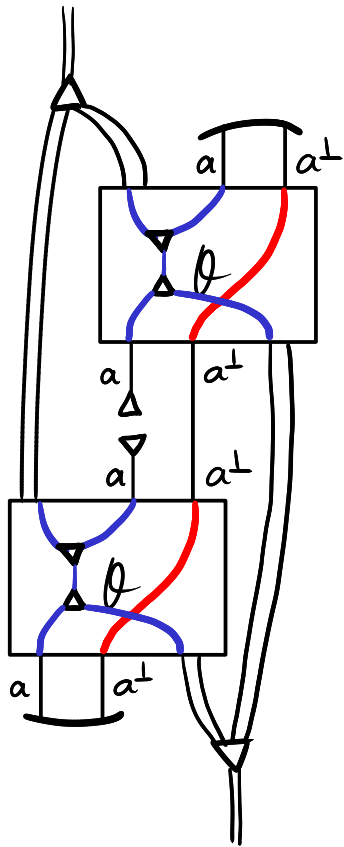
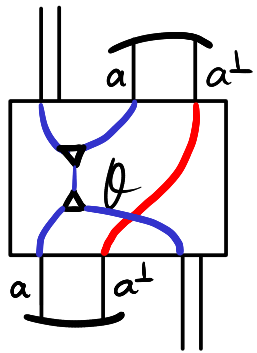
$\rightarrow_{af^*}$



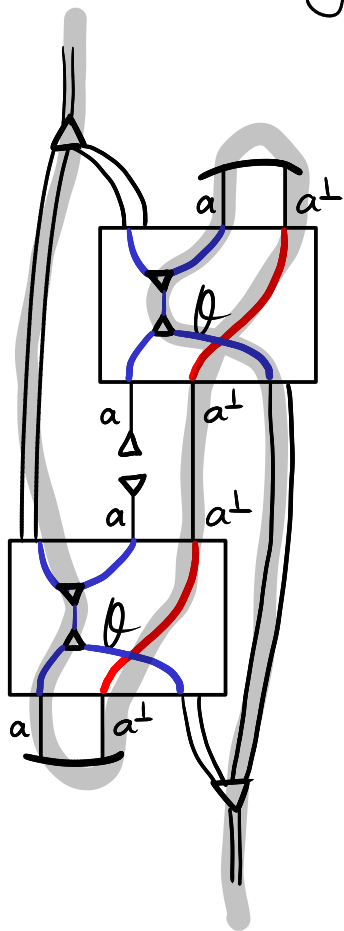
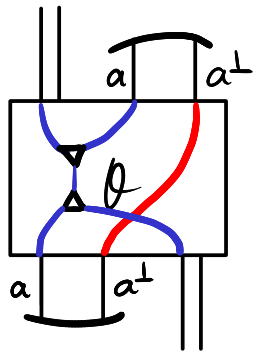
Breaking paths in atomic flows



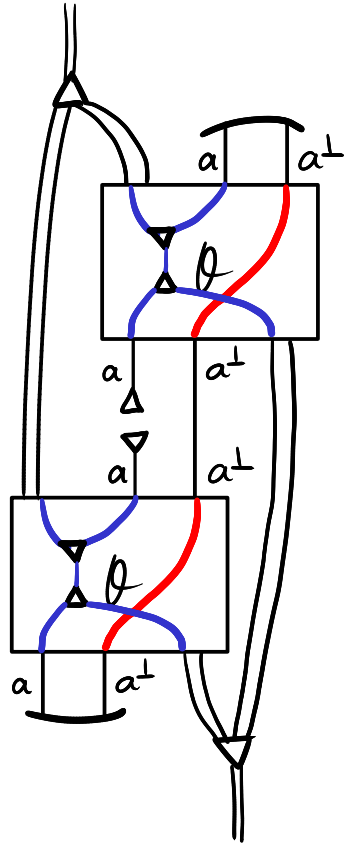
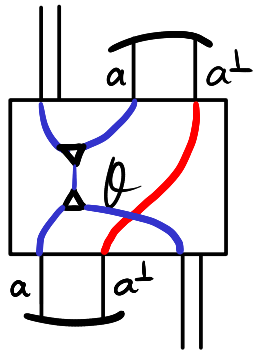
Breaking paths in atomic flows



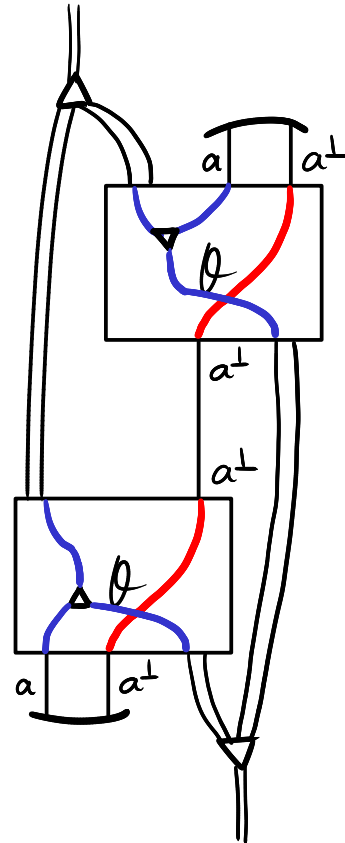
Breaking paths in atomic flows



Breaking paths in atomic flows

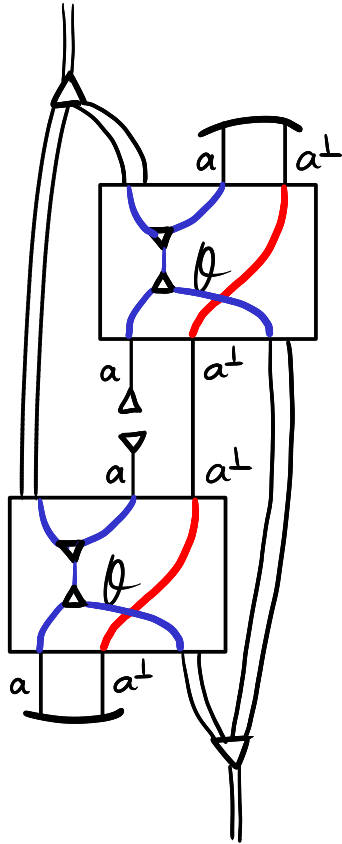
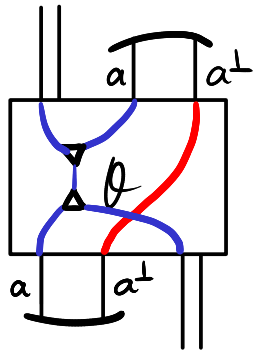


$\rightarrow_{af^*}$

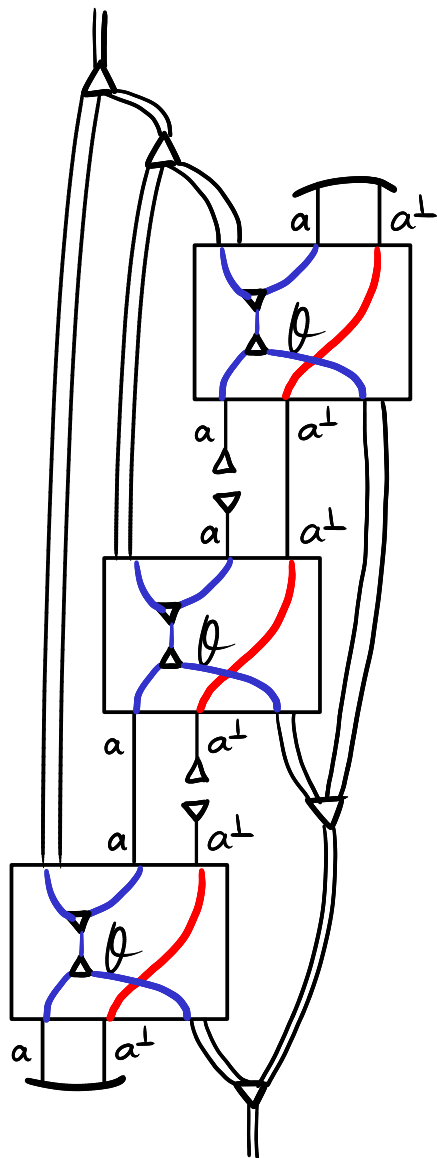
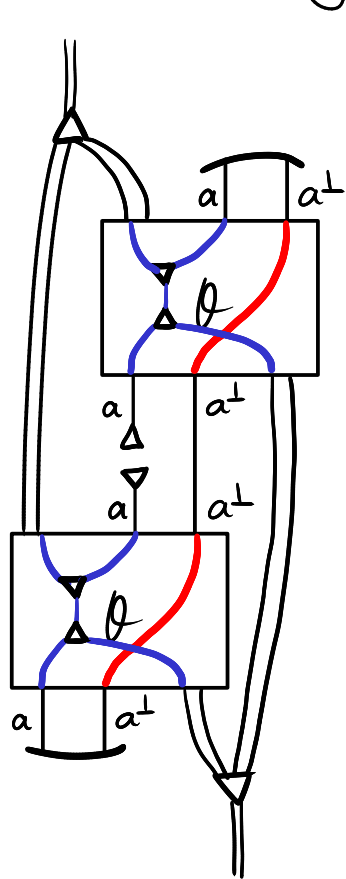
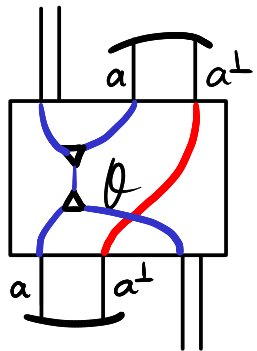


No redexes

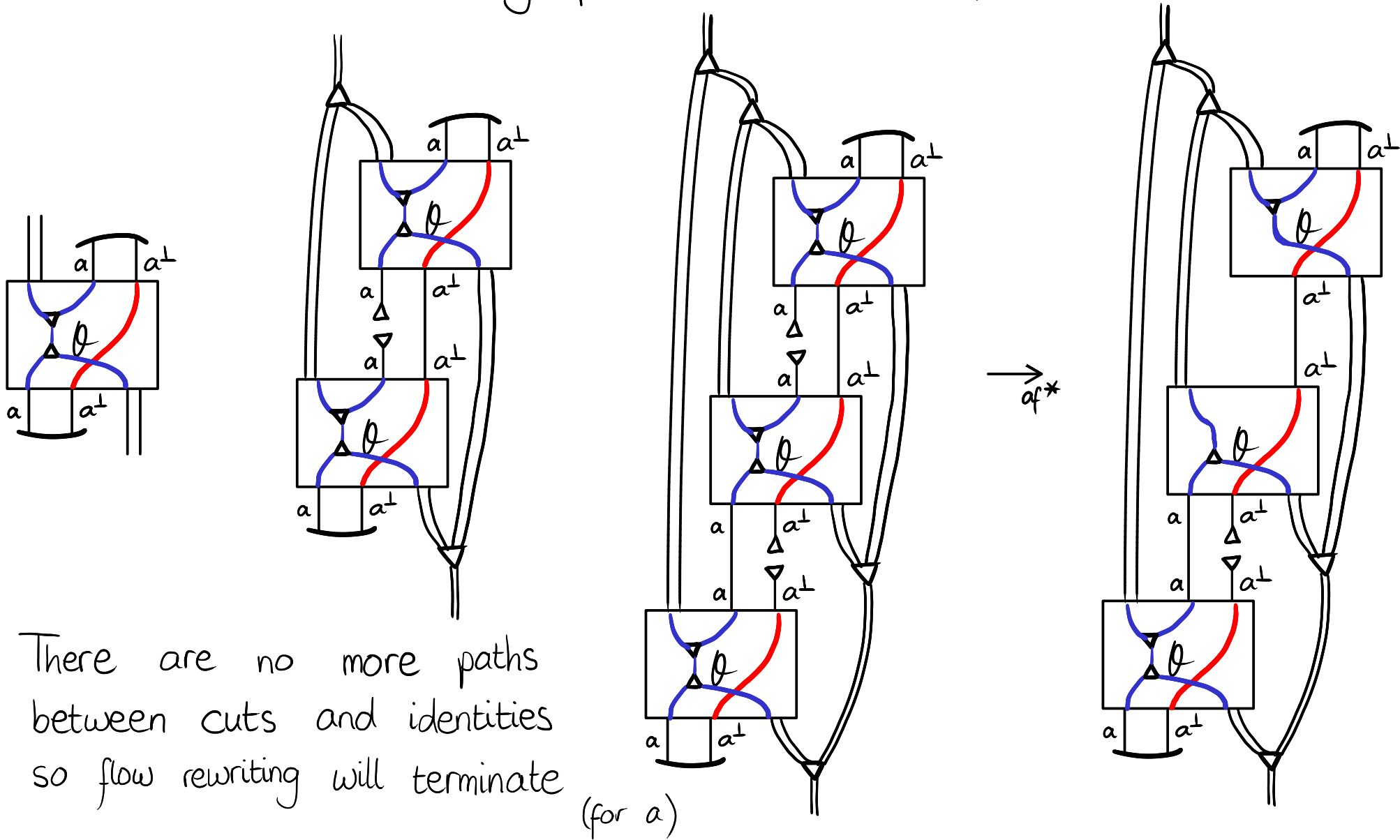
Breaking paths in atomic flows



Breaking paths in atomic flows



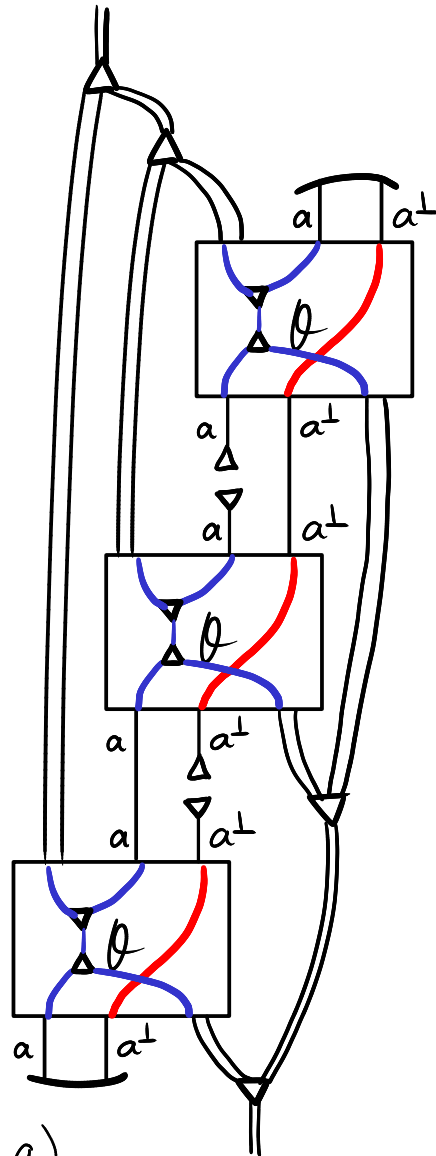
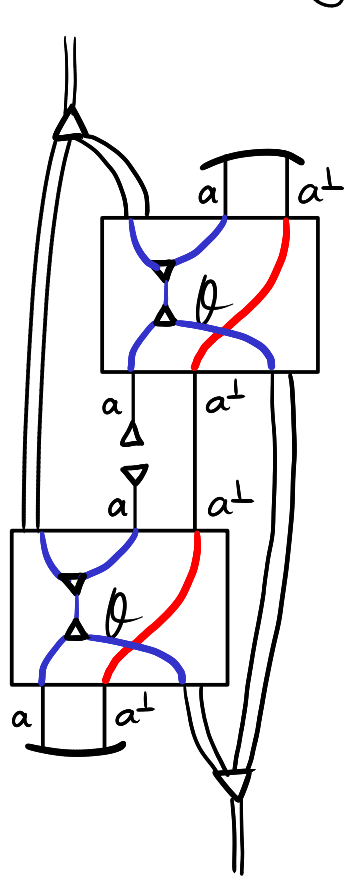
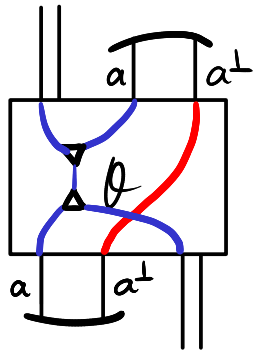
Breaking paths in atomic flows



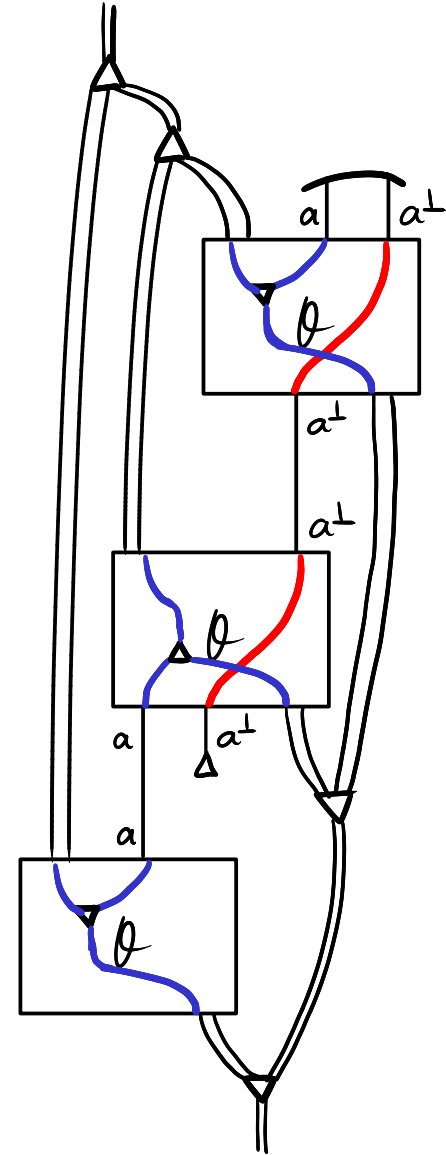
There are no more paths
between cuts and identities
so flow rewriting will terminate

(for a)

Breaking paths in atomic flows



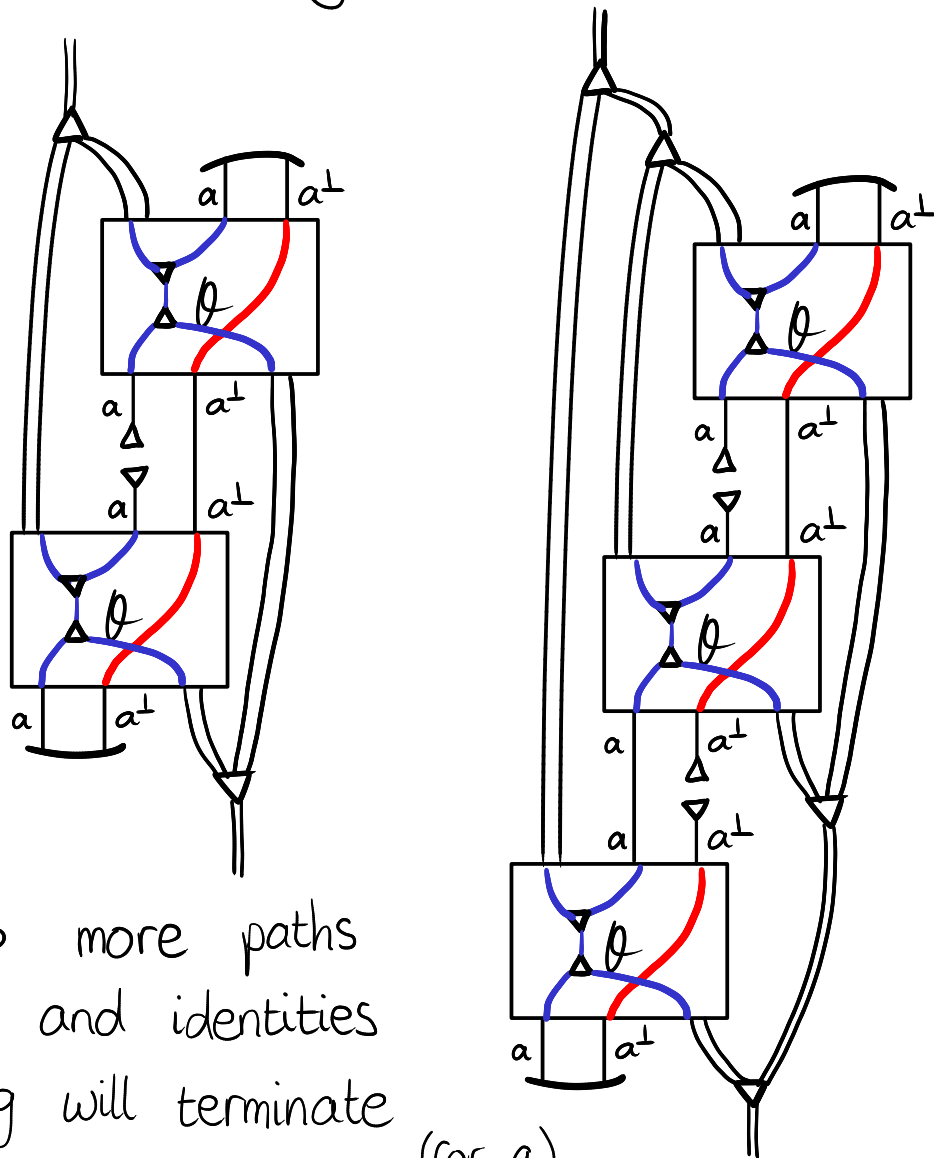
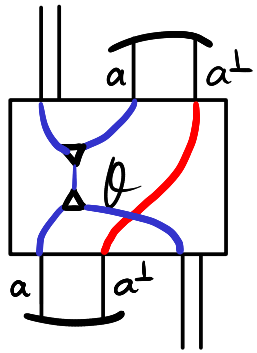
$\rightarrow_{af^*}$



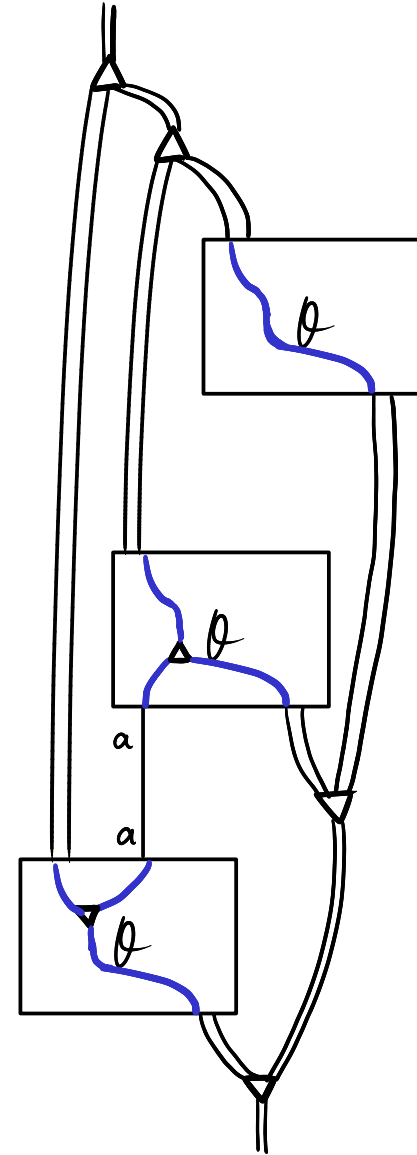
There are no more paths
between cuts and identities
so flow rewriting will terminate

(for a)

Breaking paths in atomic flows



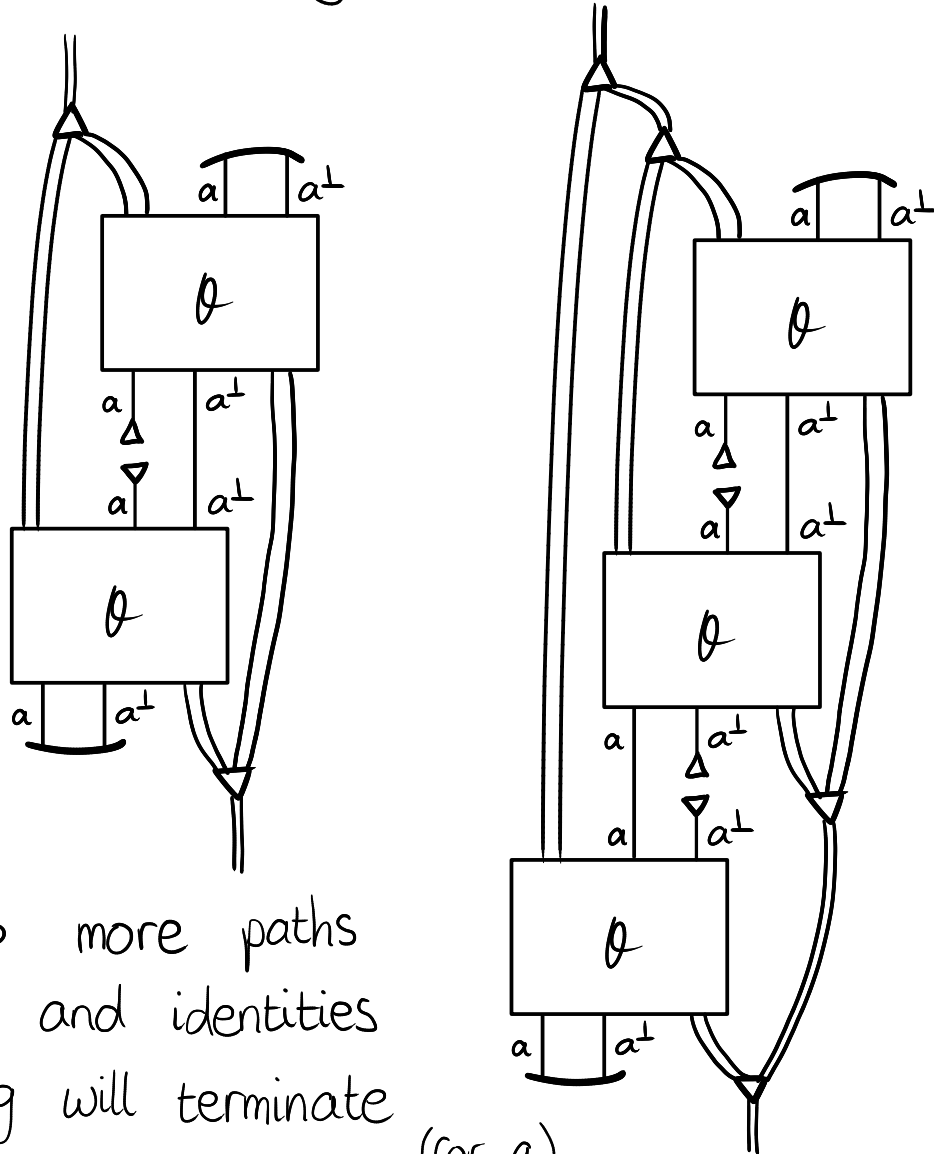
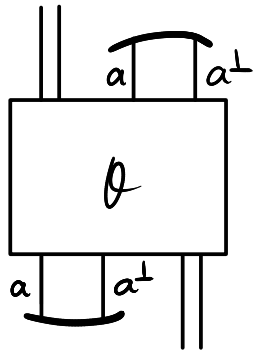
$\rightarrow_{af^*}$



There are no more paths
between cuts and identities
so flow rewriting will terminate

(for a)

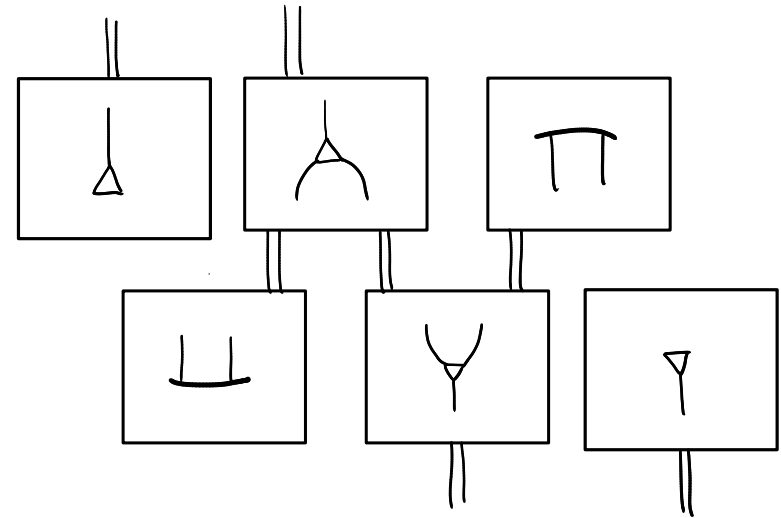
Breaking paths in atomic flows



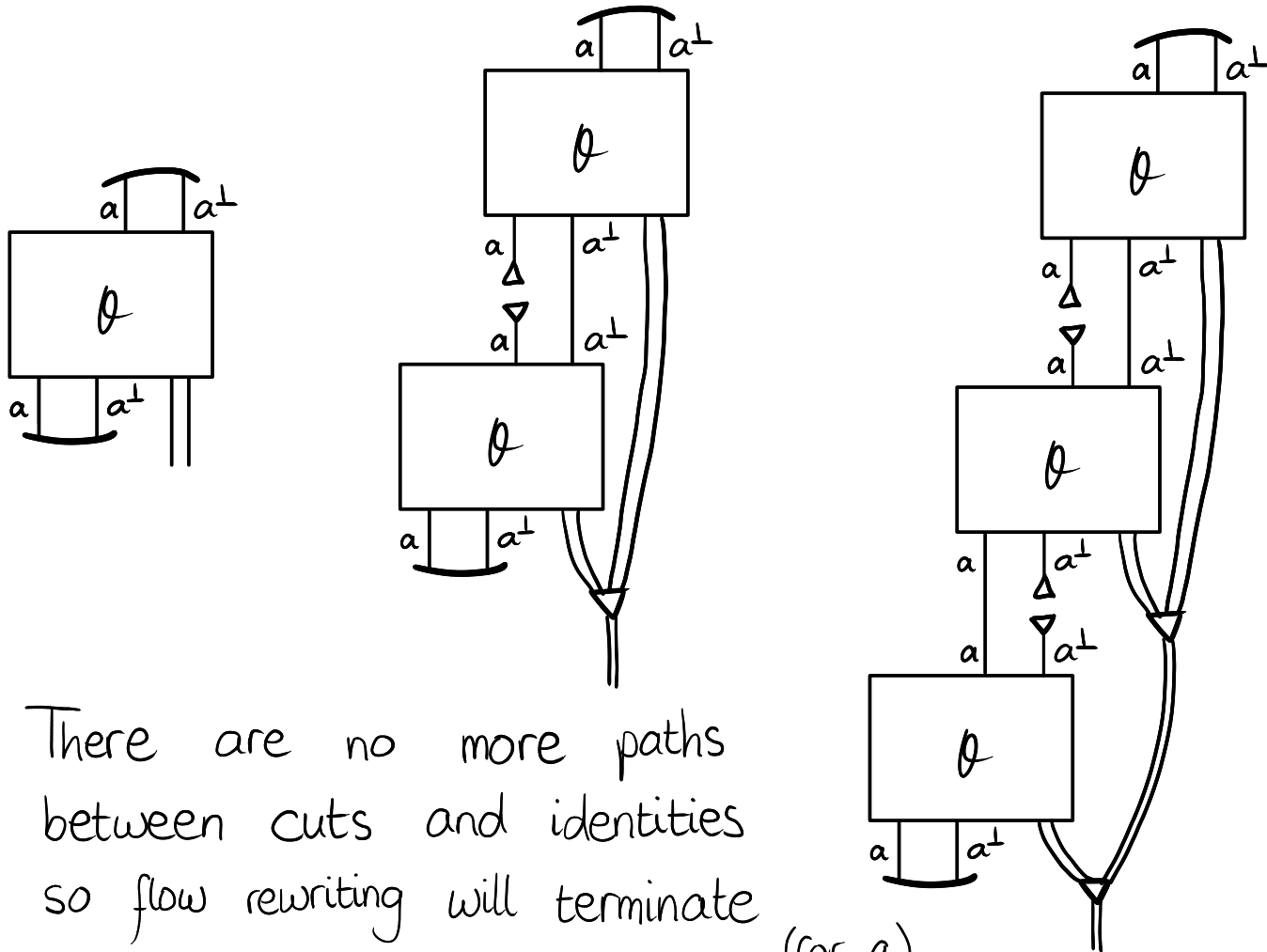
There are no more paths
between cuts and identities
so flow rewriting will terminate

(for a)

$\rightarrow_{af^*}$

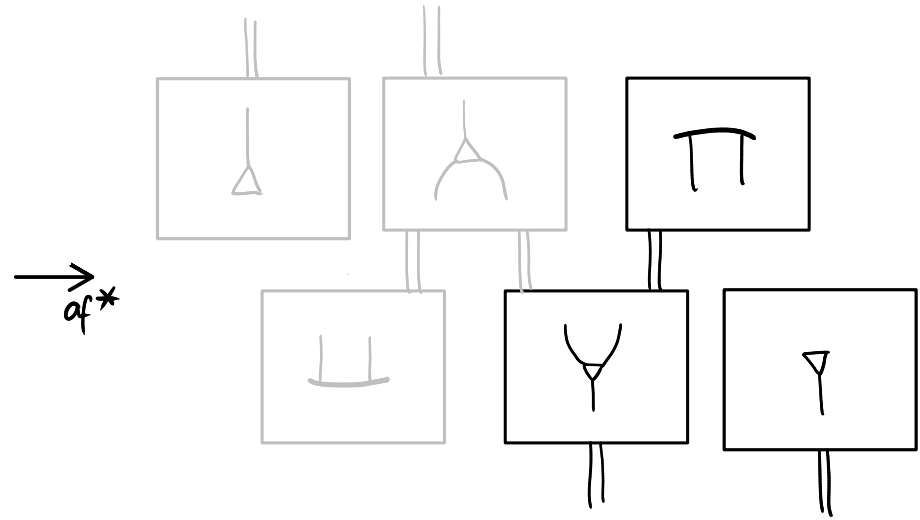


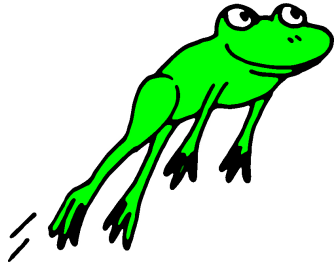
Breaking paths in atomic flows



There are no more paths between cuts and identities so flow rewriting will terminate

(for a)





Splitting and decomposition

$$i \uparrow \frac{A \otimes B \otimes (A^\perp \otimes B^\perp)}{\perp}$$

$$\xrightarrow{i \uparrow \rightarrow a i \uparrow}$$

$$= \frac{A \otimes \frac{B \otimes (A^\perp \otimes B^\perp)}{A^\perp \otimes i \uparrow \frac{B \otimes B^\perp}{\perp}}}{i \uparrow \frac{A \otimes A^\perp}{\perp}}$$

Cut cannot be made atomic
in the sequent calculus

$$\text{cut} \frac{\vdash A \otimes B, \Gamma \quad \vdash A^\perp \wp B^\perp, \Delta}{\vdash \Gamma, \Delta}$$

$$i\uparrow \frac{A \otimes B \otimes (A^\perp \wp B^\perp)}{\perp}$$

$$\xrightarrow{i\uparrow \rightarrow ai\uparrow}$$

$$= \frac{A \otimes s \frac{B \otimes (A^\perp \wp B^\perp)}{A^\perp \wp i\uparrow \frac{B \otimes B^\perp}{\perp}}}{i\uparrow \frac{A \otimes A^\perp}{\perp}}$$

Cut cannot be made atomic
in the sequent calculus

$$\text{cut} \frac{\vdash A \otimes B, \Gamma \quad \vdash A^\perp \wp B^\perp, \Delta}{\vdash \Gamma, \Delta}$$

these branches can't
be brought together

$$\text{cut} \frac{\vdash A, \Gamma \quad \text{cut} \frac{\vdash B \quad \vdash A^\perp, B^\perp, \Delta}{\vdash A^\perp, \Delta}}{\vdash \Gamma, \Delta}$$

$\vdash A \otimes B, \Gamma$

$$\text{it} \frac{A \otimes B \otimes (A^\perp \wp B^\perp)}{\perp}$$

$$\xrightarrow{\text{it} \rightarrow \text{ait}}$$

$$= \frac{A \otimes \frac{B \otimes (A^\perp \wp B^\perp)}{A^\perp \wp \text{it} \frac{B \otimes B^\perp}{\perp}}}{\text{it} \frac{A \otimes A^\perp}{\perp}}$$

Cut cannot be made atomic
in the sequent calculus

$$\text{cut} \frac{\vdash A \otimes B, \Gamma \quad \vdash A^\perp \wp B^\perp, \Delta}{\vdash \Gamma, \Delta}$$

these branches can't
be brought together

$$\text{cut} \frac{\vdash A, \Gamma \quad \text{cut} \frac{\vdash B \quad \vdash A^\perp, B^\perp, \Delta}{\vdash A^\perp, \Delta}}{\vdash \Gamma, \Delta}$$

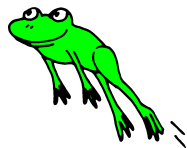
$\vdash A \otimes B, \Gamma$

Splitting of the context is
guaranteed by the formalism
in the sequent calculus, but
not in deep inference

$$i\uparrow \frac{A \otimes B \otimes (A^\perp \wp B^\perp)}{\perp}$$

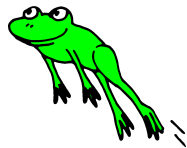
$$\xrightarrow{i\uparrow \rightarrow ai\uparrow}$$

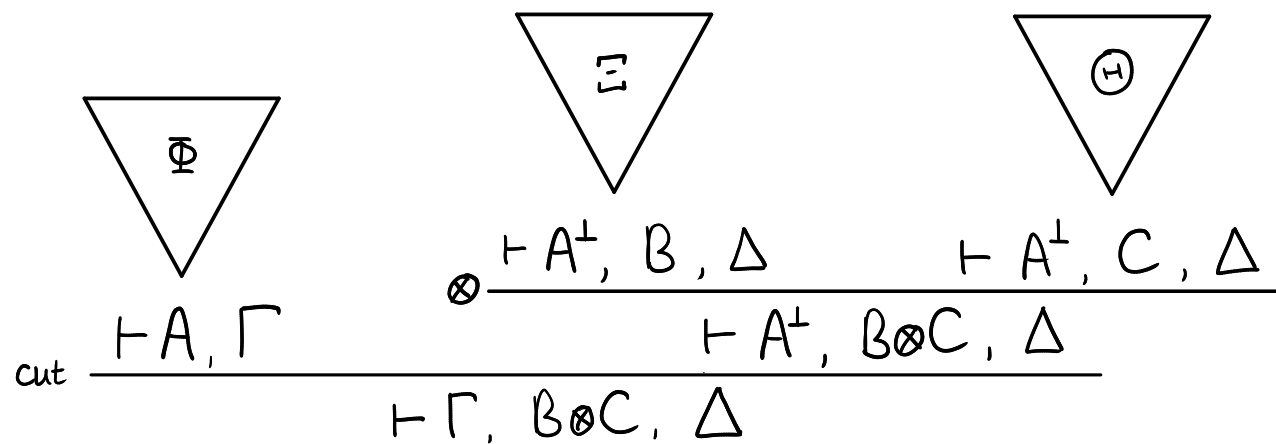
$$= \frac{A \otimes s \frac{B \otimes (A^\perp \wp B^\perp)}{A^\perp \wp i\uparrow \frac{B \otimes B^\perp}{\perp}}}{i\uparrow \frac{A \otimes A^\perp}{\perp}}$$



$$\begin{array}{c}
 \text{cut} \frac{
 \begin{array}{c}
 \triangle \Phi \\
 \hline
 \vdash A, \Gamma
 \end{array}
 \quad
 \begin{array}{c}
 \triangle \exists \\
 \hline
 \otimes \frac{
 \vdash A^\perp, B, \Delta \quad \vdash A^\perp, C, \Delta
 }{
 \vdash A^\perp, B \otimes C, \Delta
 }
 \end{array}
 }{
 \vdash \Gamma, B \otimes C, \Delta
 }
 \end{array}$$

Splitting of the context is
 guaranteed by the formalism
 in the sequent calculus, but
 not in deep inference

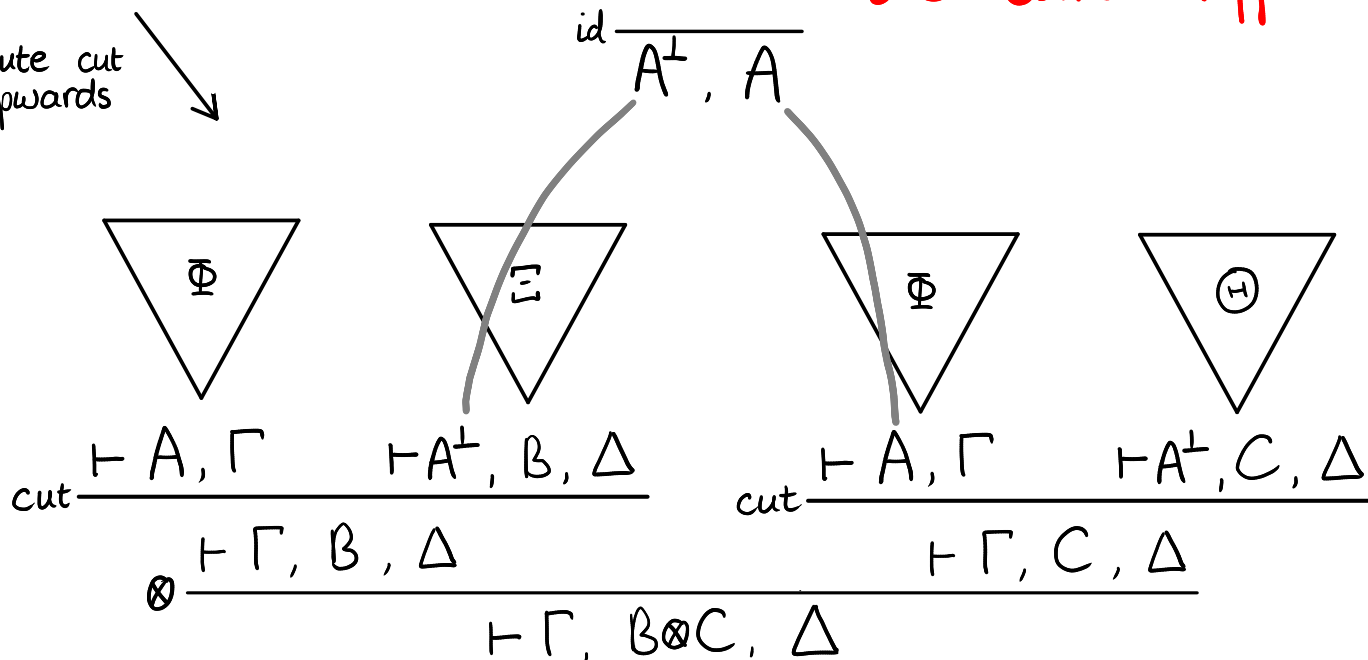




permute cut upwards

this can't happen

Splitting of the context is guaranteed by the formalism in the sequent calculus, but not in deep inference



$$\begin{array}{c}
 \text{a} \downarrow \frac{}{\top} \\
 \text{a} \uparrow \frac{a}{a \wedge a} \quad \vee \text{a} \uparrow \frac{a^\perp}{a^\perp \wedge a^\perp} \\
 \text{m} \frac{}{(a \vee a^\perp) \wedge (a \vee a^\perp)} \\
 \text{2s} \frac{}{a \wedge a^\perp \vee a^\perp \vee a} \\
 \text{a} \uparrow \frac{}{\perp}
 \end{array}$$

Splitting of the context is guaranteed by the formalism in the sequent calculus, but not in deep inference



this can't happen

$$\begin{array}{c}
 \text{id} \frac{}{A^\perp, A} \\
 \swarrow \quad \searrow \\
 \begin{array}{c}
 \triangle \Phi \\
 \hline
 \text{cut} \frac{\vdash A, \Gamma}{\vdash \Gamma, B, \Delta}
 \end{array}
 \quad
 \begin{array}{c}
 \triangle \Xi \\
 \hline
 \text{cut} \frac{\vdash A^\perp, B, \Delta}{\vdash \Gamma, B, \Delta}
 \end{array}
 \quad
 \begin{array}{c}
 \triangle \Phi \\
 \hline
 \text{cut} \frac{\vdash A, \Gamma}{\vdash \Gamma, C, \Delta}
 \end{array}
 \quad
 \begin{array}{c}
 \triangle \ominus \\
 \hline
 \text{cut} \frac{\vdash A^\perp, C, \Delta}{\vdash \Gamma, C, \Delta}
 \end{array}
 \end{array}$$

$$\otimes \frac{\vdash \Gamma, B, \Delta \quad \vdash \Gamma, C, \Delta}{\vdash \Gamma, B \otimes C, \Delta}$$

LS

$$c \downarrow \frac{A \oplus A}{A} \quad \leftrightarrow \frac{0}{A} \quad d \downarrow \frac{(A \wp B) \wp (C \wp D)}{(A \wp C) \wp (B \oplus D)}$$

MAV

$$m \downarrow \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)}$$

MALS

MLS

$$a \downarrow \frac{1}{a \wp a^\perp} \quad s \frac{A \oplus (B \wp C)}{(A \oplus B) \wp C}$$

$$q \downarrow \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)}$$

BU

$$w \downarrow \frac{1}{?A} \quad g \downarrow \frac{??A}{?A} \quad b \downarrow \frac{?A \wp A}{?A} \quad p \downarrow \frac{!(A \wp B)}{!A \wp ?B} \quad e \downarrow \frac{1}{!1}$$

NEL

$$\text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

This example is to show the *simplest possible* case of a cut in an MLS+cut proof, it is not intended as a general case, but a best case scenario to show what is intended by the splitting procedure



$$\text{ai} \uparrow \frac{a \otimes a^\perp}{\perp} \quad \wp \quad \Pi \quad B$$

$$\text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

Π

$$\text{ai} \uparrow \frac{a \otimes a^\perp}{\perp} \quad \wp \quad \begin{array}{c} K\{a^\perp\} \wp H\{a\} \\ \Delta \parallel \\ B \end{array}$$

$$\text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

$$=, 2s \frac{\begin{array}{c} \top \\ a \wp K\{a^\perp\} \end{array} \otimes \begin{array}{c} \top \\ a^\perp \wp H\{a\} \end{array}}{\begin{array}{c} \text{ai} \uparrow \frac{a \otimes a^\perp}{\perp} \wp \begin{array}{c} K\{a^\perp\} \wp H\{a\} \\ \Delta \parallel \\ \mathcal{B} \end{array} \end{array}}$$

$$a \downarrow \frac{1}{a \otimes a^\perp} \quad s \frac{A \otimes (B \otimes C)}{(A \otimes B) \otimes C}$$

MLS

$$=, 2s \frac{\begin{array}{c} \top \\ \rho \otimes a^\perp \\ a \otimes \Omega \parallel \\ K\{a^\perp\} \end{array} \otimes \begin{array}{c} \top \\ a^\perp \otimes H\{a\} \end{array}}{\begin{array}{c} a \otimes a^\perp \\ a \uparrow \frac{}{\perp} \end{array} \otimes \begin{array}{c} K\{a^\perp\} \otimes H\{a\} \\ \Delta \parallel \\ B \end{array}}$$

$$\text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

$$\begin{array}{c} \pi \top \\ \text{ai} \downarrow \frac{1}{a \wp a^\perp} \otimes \rho \\ \text{=, s} \frac{\rho \otimes a^\perp}{a \wp \Omega \parallel K\{a^\perp\}} \otimes \frac{\top}{a^\perp \wp H\{a\}} \\ \text{=, 2s} \frac{a \otimes a^\perp}{\perp} \wp \frac{K\{a^\perp\} \wp H\{a\}}{\Delta \parallel B} \end{array}$$

$$\text{ail} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

In this best case scenario, we are able to extract the pieces of the proof that are necessary to rebuild a cut-free proof of the same conclusion



$$\begin{array}{c} \pi \top \\ \text{ail} \frac{1}{a \wp a^\perp} \otimes p \\ \text{=, s} \frac{}{p \otimes a^\perp} \\ a \wp \Omega \parallel \\ K\{a^\perp\} \end{array} \quad \otimes \quad \begin{array}{c} \psi \top \\ \text{ail} \frac{1}{a^\perp \wp a} \otimes Q \\ \text{s} \frac{}{a \otimes Q} \\ a^\perp \wp \Theta \parallel \\ H\{a\} \end{array}$$

$$\text{=, 2s} \frac{}{a \otimes a^\perp \wp K\{a^\perp\} \wp H\{a\}}$$

$$\text{ail} \frac{a \otimes a^\perp}{\perp} \quad \wp \quad \begin{array}{c} K\{a^\perp\} \wp H\{a\} \\ \Delta \parallel \\ \beta \end{array}$$

→

$$\begin{array}{c} \pi \top \otimes \text{ail} \frac{1}{a^\perp \wp a} \otimes \psi \top \\ p \otimes a^\perp \otimes Q \\ \text{2s} \frac{}{p \otimes a^\perp \wp a \otimes Q} \\ \Omega \parallel \quad \wp \quad \Theta \parallel \\ K\{a^\perp\} \quad \wp \quad H\{a\} \\ \Delta \parallel \\ \beta \end{array}$$

$$\text{ail} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

If $(A \otimes B) \wp C \stackrel{\Pi_{\text{MLS}}}{\sim}$ then there exist K_A, K_B such that

$$K_A \wp K_B \stackrel{\Pi_{\text{MLS}}}{\sim} K, \quad A \wp K_A, \quad B \wp K_B$$

If $a \wp C \stackrel{\Pi_{\text{MLS}}}{\sim}$ then $a^\perp \stackrel{\Pi_{\text{MLS}}}{\sim} C$



This is the theorem that we need to prove in order to extend the technique to more general cases

$$\begin{array}{c} \Pi \Pi \\ \text{ail} \frac{1}{a \wp a^\perp} \otimes p \\ \text{=, s} \frac{}{p \otimes a^\perp} \\ a \wp \Omega \parallel \\ K\{a^\perp\} \end{array} \quad \otimes \quad \begin{array}{c} \Psi \Pi \\ \text{ail} \frac{1}{a^\perp \wp a} \otimes Q \\ \text{s} \frac{}{a \otimes Q} \\ a^\perp \wp \Theta \parallel \\ H\{a\} \end{array}$$

$$\text{=, 2s} \frac{}{a \otimes a^\perp \wp K\{a^\perp\} \wp H\{a\}} \quad \Delta \parallel \quad \perp$$

→

$$\begin{array}{c} \Pi \Pi \otimes \text{ail} \frac{1}{a^\perp \wp a} \otimes \Psi \Pi \\ p \otimes a^\perp \otimes Q \\ \Omega \parallel \wp \Theta \parallel \\ K\{a^\perp\} \wp H\{a\} \\ \Delta \parallel \\ \perp \end{array}$$

$$\text{ai} \downarrow \frac{1}{a \otimes a^\perp} \quad \text{s} \frac{A \otimes (B \otimes C)}{(A \otimes B) \otimes C}$$

MLS

If $(A \otimes B) \otimes C \stackrel{\Pi_{\text{MLS}}}{\sim}$ then there exist K_A, K_B such that

$$K_A \otimes K_B \stackrel{\Pi_{\text{MLS}}}{\sim} K, \quad A \otimes K_A \stackrel{\Pi_{\text{MLS}}}{\sim}, \quad B \otimes K_B \stackrel{\Pi_{\text{MLS}}}{\sim}$$

If $a \otimes C \stackrel{\Pi_{\text{MLS}}}{\sim}$ then $a^\perp \stackrel{\Pi_{\text{MLS}}}{\sim} C$



$$\begin{aligned} & \stackrel{\Pi_{\text{MLS}}}{\sim} \\ & \stackrel{=, 2s}{=} \frac{(a \otimes K\{a^\perp\}) \otimes (a^\perp \otimes H\{a\})}{(a \otimes a^\perp) \otimes K\{a^\perp\} \otimes H\{a\}} \otimes 0 \\ & = \frac{\text{ai} \uparrow \frac{a \otimes a^\perp}{\perp} \otimes K\{a^\perp\} \otimes H\{a\} \otimes 0}{\Delta \parallel B} \end{aligned}$$

splitting
→

$$\begin{aligned} & \frac{\Pi}{(a \otimes K\{a^\perp\} \otimes D_1)} \otimes \frac{\Pi}{(a^\perp \otimes H\{a\} \otimes D_2)} \\ & \frac{((a \otimes K\{a^\perp\}) \otimes (a^\perp \otimes H\{a\})) \otimes D_1 \otimes D_2}{\parallel 0} \end{aligned}$$

$$\text{ax} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

If $(A \otimes B) \wp C \stackrel{\Pi_{\text{MLS}}}{\vdash}$ then there exist K_A, K_B such that $K_A \wp K_B \stackrel{\Pi_{\text{MLS}}}{\vdash} K$, $A \wp K_A \stackrel{\Pi_{\text{MLS}}}{\vdash}$, $B \wp K_B \stackrel{\Pi_{\text{MLS}}}{\vdash}$

If $a \wp C \stackrel{\Pi_{\text{MLS}}}{\vdash}$ then $a^\perp \stackrel{\Pi_{\text{MLS}}}{\vdash} C$



Proof is by induction on $(|(A \otimes B) \wp C|, n(\Phi))$, where $|A|$ is the number of atom occurrences in A and $n(\Phi)$ is the number of inference steps in Φ .

Cases : (i) $\Phi = \frac{\Phi' \Pi}{\text{r} \frac{A'}{A} \otimes B} \wp C$ (ii) $\Phi = \frac{\Phi' \Pi}{A \otimes \text{r} \frac{B'}{B}} \wp C$ (iii) $\Phi = \frac{\Phi' \Pi}{(A \otimes B) \wp \text{r} \frac{C'}{C}}$

(iv) $\Phi = \text{s} \frac{\frac{\Phi' \Pi}{A_1 \otimes B_1 \otimes ((A_2 \otimes B_2) \wp C_1)}}{(A_1 \otimes A_2 \otimes B_1 \otimes B_2) \wp C_1} \wp C_2$

(v) $\Phi = \frac{\frac{\Phi' \Pi}{((A \otimes B) \wp C_1 \wp C_2) \otimes C_3}}{(A \otimes B) \wp C_1 \wp (C_2 \otimes C_3)} \wp C_4$

If $(A \otimes B) \wp C$ then there exist K_A, K_B such that $K_A \wp K_B \parallel_{MLS}^K$, $A \wp K_A$, $B \wp K_B$

$$(iv) \quad \Phi = \frac{\Phi' \Pi}{(A_1 \otimes A_2 \otimes B_1 \otimes B_2) \wp C_1} \wp C_2$$

Observation : For all $\frac{A}{B}$ in MLS , $|A| \leq |B|$

By induction on Φ' , there exist D, E s.t. $D \wp E \parallel_{MLS}^{D \wp E}$, $(A_1 \otimes B_1) \wp D$, $(A_2 \otimes B_2) \wp C_1 \wp E$

By induction on Θ , there exist D_1, D_2 s.t. $D_1 \wp D_2 \parallel_{MLS}^{D_1 \wp D_2}$, $A_1 \wp D_1$, $B_1 \wp D_2$

By induction on Ω , there exist E_1, E_2 s.t. $E_1 \wp E_2 \parallel_{MLS}^{E_1 \wp E_2}$, $A_2 \wp E_1$, $B_2 \wp E_2$

Then we take $K_A = D_1 \wp E_1$, $K_B = D_2 \wp E_2$

$$\frac{\frac{\Pi_{MLS}}{A_1 \wp D_1} \otimes \frac{\Pi_{MLS}}{A_2 \wp E_1}}{\frac{\Pi_{MLS}}{(A_1 \otimes A_2) \wp D_1 \wp E_1}} \quad \frac{\frac{\Pi_{MLS}}{B_1 \wp D_2} \otimes \frac{\Pi_{MLS}}{B_2 \wp E_2}}{\frac{\Pi_{MLS}}{(B_1 \otimes B_2) \wp D_2 \wp E_2}}$$

$$a_i \downarrow \frac{1}{a \otimes a^\perp} \quad s \frac{A \otimes (B \otimes C)}{(A \otimes B) \otimes C}$$

MLS

$\Phi \parallel_{MLS}$

$$\left(a_i \uparrow \frac{a \otimes a^\perp}{\perp} \otimes B \right) \otimes C$$

$$\text{ai} \frac{1}{a \otimes a^\perp} \quad \text{s} \frac{A \otimes (B \otimes C)}{(A \otimes B) \otimes C}$$

MLS

$$\Phi \parallel_{MLS} \left(\text{ai} \frac{a \otimes a^\perp}{\perp} \otimes B \right) \otimes C$$

splitting on Φ



$$\begin{array}{c} \Psi \parallel_{MLS} \\ \text{ai} \frac{a \otimes a^\perp}{\perp} \otimes C_1 \quad \otimes \quad B \otimes C_2 \\ \hline \text{2s} \quad \quad \quad C_1 \otimes C_2 \\ \quad \quad \quad \parallel \\ \quad \quad \quad C \end{array}$$

$$\text{ai} \frac{1}{a \otimes a^\perp} \quad \text{s} \frac{A \otimes (B \otimes C)}{(A \otimes B) \otimes C}$$

MLS

$$\Phi \prod_{MLS} \left(\text{ai} \frac{a \otimes a^\perp}{\perp} \otimes B \right) \otimes C$$

splitting on Φ

$$\Psi \prod_{MLS} \text{ai} \frac{a \otimes a^\perp}{\perp} \otimes C_1 \otimes \prod_{MLS} B \otimes C_2$$

$$2s \quad (\perp \otimes B) \otimes C_1 \otimes C_2$$

$$\parallel$$

$$C$$

splitting on Ψ

$$\otimes \prod_{MLS} a \otimes D_1 \otimes \otimes \prod_{MLS} a^\perp \otimes D_2$$

$$\frac{a \otimes a^\perp}{\perp} \otimes D_1 \otimes D_2 \otimes \prod B \otimes C_2$$

$$\parallel$$

$$C_1$$

$$2s \quad (\perp \otimes B) \otimes C_1 \otimes C_2$$

$$\parallel_{MLS}$$

$$C$$

$$\text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C}$$

MLS

This example shows how we iterate splitting in order to eliminate even a deeply nested cut.

$$\Phi \prod_{MLS}$$

$$\left(\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \otimes B \right) \wp C$$

splitting on Φ

$$\Psi \prod_{MLS}$$

$$\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \wp C_1 \otimes \prod_{MLS} B \wp C_2$$

$$2s \frac{(\perp \otimes B) \wp C_1 \wp C_2}{C}$$

splitting on Ψ

$$\otimes \prod_{MLS} a \wp D_1 \otimes \otimes \prod_{MLS} a^\perp \wp D_2$$

$$\frac{a \otimes a^\perp}{\perp} \wp D_1 \wp D_2 \otimes \prod B \wp C_2$$

$$2s \frac{(\perp \otimes B) \wp C_1 \wp C_2}{C}$$

atomic splitting
on Θ and Ω

$$\text{ai} \downarrow \frac{1}{a^\perp \quad a}$$

$$\parallel_{MLS} \wp D_1 \quad \parallel_{MLS} \wp D_2 \otimes \prod B \wp C_2$$

$$2s \frac{C_1 \wp C_2}{C}$$

$$a_i \downarrow \frac{1}{a \otimes a^\perp} \quad s \frac{A \otimes (B \otimes C)}{(A \otimes B) \otimes C}$$

MLS

What about another system?



$$\Phi \prod_{MLS} \left(a_i \downarrow \frac{a \otimes a^\perp}{\perp} \otimes B \right) \otimes C$$

splitting on Φ

$$\Psi \prod_{MLS} \quad a_i \downarrow \frac{a \otimes a^\perp}{\perp} \otimes C_1 \otimes \prod_{MLS} B \otimes C_2$$

$$2s \quad (\perp \otimes B) \otimes C_1 \otimes C_2$$

$$\parallel C$$

splitting on Ψ

$$\otimes \prod_{MLS} a \otimes D_1 \otimes \otimes \prod_{MLS} a^\perp \otimes D_2$$

$$\frac{a \otimes a^\perp}{\perp} \otimes D_1 \otimes D_2 \otimes \prod B \otimes C_2$$

$$2s \quad (\perp \otimes B) \otimes C_1 \otimes C_2$$

$$\parallel_{MLS} C$$

atomic splitting on Θ and Ω

$$a_i \downarrow \frac{1}{a^\perp \parallel_{MLS} D_1 \otimes a \parallel_{MLS} D_2} \otimes \prod B \otimes C_2$$

$$2s \quad C_1 \otimes C_2$$

$$(\perp \otimes B) \otimes \parallel_{MLS} C$$

$$\begin{array}{l} \text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{qv} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \end{array}$$

BU

What about another system?



$$\Phi \prod_{BU} \left(\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \triangleleft B \right) \wp C$$

splitting on Φ

$$\Psi \prod_{BU} \frac{\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \wp C_1 \triangleleft B \wp C_2}{2s \frac{(\perp \triangleleft B) \wp C_1 \triangleleft C_2}{C} \parallel_{BU} C}$$

splitting on Ψ

$$\begin{array}{l} \Theta \prod_{BU} \frac{a \wp D_1}{\perp} \otimes \Omega \prod_{BU} \frac{a^\perp \wp D_2}{\perp} \triangleleft B \wp C_2 \\ \frac{a \otimes a^\perp}{\perp} \wp \frac{D_1 \wp D_2}{\parallel_{BU} C_1} \end{array} \xrightarrow{2s} \frac{(\perp \triangleleft B) \wp C_1 \triangleleft C_2}{C} \parallel_{BU} C$$

atomic splitting on Θ and Ω

$$\begin{array}{l} \text{ai} \downarrow \frac{1}{a^\perp \parallel_{BU} D_1 \wp a \parallel_{BU} D_2} \triangleleft B \wp C_2 \\ \parallel_{BU} C_1 \end{array} \xrightarrow{2s} \frac{(\perp \triangleleft B) \wp C_1 \triangleleft C_2}{C} \parallel_{BU} C$$

$$\text{ax} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{qv} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)}$$

BV

If $\Phi \vdash_{BV} (A \otimes B) \wp C$ then there exist K_A, K_B such that

$$K_A \wp K_B \quad \frac{}{K} \parallel_{BV}, \quad A \wp K_A, \quad B \wp K_B$$

If $\Phi \vdash_{BV} (A \triangleleft B) \wp C$ then there exist K_A, K_B such that

$$K_A \triangleleft K_B \quad \frac{}{K} \parallel_{BV}, \quad A \wp K_A, \quad B \wp K_B$$

If $\vdash_{BV} a \wp C$ then $\frac{}{C} \parallel_{BV} a^\perp$



This is the theorem that we need to prove in order to extend the splitting technique to system BV+cut

Identifying and proving the cases left as an exercise

Cases : (i) $\Phi = \frac{\Phi' \vdash (A_1 \wp C_1) \triangleleft ((A_2 \triangleleft B) \wp C_2)}{\text{qv} \frac{(A_1 \triangleleft A_2 \triangleleft B) \wp (C_1 \triangleleft C_2)}} \wp C_3$

(ii) $\Phi = ?$...

$$\begin{array}{c}
 \text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \downarrow \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \downarrow \frac{\perp}{?A} \quad \text{g} \downarrow \frac{??A}{?A} \quad \text{b} \downarrow \frac{?A \wp A}{?A} \quad \text{p} \downarrow \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \downarrow \frac{1}{!1}
 \end{array}$$

NEL

$$\begin{array}{c}
 \Phi \parallel \text{NEL} \\
 ! \left(\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \right) \wp C
 \end{array}$$

What about
another
system?..



$$\begin{array}{c}
 \text{ai} \downarrow \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \downarrow \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \downarrow \frac{\perp}{?A} \quad \text{g} \downarrow \frac{??A}{?A} \quad \text{b} \downarrow \frac{?A \wp A}{?A} \quad \text{p} \downarrow \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \downarrow \frac{1}{!1}
 \end{array}$$

$$! \left(\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \right) \wp C \quad \Phi \parallel \text{NEL}$$

splitting
on Φ

$$\begin{array}{c}
 ! \left(\text{ai} \downarrow \frac{a \otimes a^\perp}{\perp} \wp C_1 \wp \dots \wp C_n \right) \quad \Psi \parallel \text{NEL} \\
 \text{p} \downarrow \frac{}{!(\perp \wp C_1 \wp \dots \wp C_{n-1}) \wp ?C_n} \\
 \text{p} \downarrow \frac{}{\vdots} \\
 \text{p} \downarrow \frac{}{!\perp \wp \left(?C_1 \wp \dots \wp ?C_n \right)} \\
 \parallel \text{NEL} \\
 C
 \end{array}$$

splitting on Ψ
and atomic
splitting

$$\begin{array}{c}
 \text{e} \downarrow \frac{1}{\left(\begin{array}{c} \text{ai} \downarrow \frac{1}{\begin{array}{c} a^\perp \quad a \\ \parallel \text{NEL} \wp \parallel \text{NEL} \\ D_1 \quad D_2 \end{array}} \\ \parallel \\ C_1 \wp \dots \wp C_n \\ = \frac{}{\perp \wp C_1 \wp \dots \wp C_n} \end{array} \right)} \\
 \parallel \text{p} \downarrow \\
 !\perp \wp \left(?C_1 \wp \dots \wp ?C_n \right) \\
 \parallel \text{NEL} \\
 C
 \end{array}$$

What about
another
system?..



NEL

$$\begin{array}{c}
 \text{ax} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \frac{\perp}{?A} \quad \text{g} \frac{??A}{?A} \quad \text{b} \frac{?A \wp A}{?A} \quad \text{p} \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \frac{1}{!1}
 \end{array}$$

If $\frac{\perp \Pi_{NEL}}{(A \otimes B) \wp C}$ then there exist K_A, K_B such that

$$\begin{array}{c}
 K_A \wp K_B \\
 \parallel_{NEL} \\
 K
 \end{array}, \quad \frac{\Pi_{NEL}}{A \wp K_A}, \quad \frac{\Pi_{NEL}}{B \wp K_B}$$

If $\frac{\perp \Pi_{NEL}}{(A \triangleleft B) \wp C}$ then there exist K_A, K_B such that

$$\begin{array}{c}
 K_A \triangleleft K_B \\
 \parallel_{NEL} \\
 K
 \end{array}, \quad \frac{\Pi_{NEL}}{A \wp K_A}, \quad \frac{\Pi_{NEL}}{B \wp K_B}$$

If $\frac{\perp \Pi_{NEL}}{!A \wp C}$ then there exist $K_1, \dots, K_n$ such that

$$\begin{array}{c}
 ?K_1 \wp \dots \wp ?K_n \\
 \parallel_{NEL} \\
 C
 \end{array}, \quad \frac{\Pi_{NEL}}{A \wp K_1 \wp \dots \wp K_n}$$

If $\frac{\perp \Pi_{NEL}}{?A \wp C}$ then there exists K such that

$$\begin{array}{c}
 !K \\
 \parallel_{NEL} \\
 C
 \end{array}, \quad \frac{\Pi_{NEL}}{A \wp K}$$

If $\frac{\Pi_{NEL}}{a \wp C}$ then $\frac{a^\perp}{\parallel_{NEL} C}$

NEL

$$\text{ax} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \frac{\perp}{?A} \quad \text{g} \frac{??A}{?A} \quad \text{b} \frac{?A \wp A}{?A} \quad \text{p} \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \frac{1}{!1}$$

If $\frac{\perp}{(A \otimes B) \wp C} \Pi_{NEL}$ then there exist K_A, K_B such that

$$\frac{K_A \wp K_B}{K} \Pi_{NEL}, \quad \frac{\Pi_{NEL}}{A \wp K_A}, \quad \frac{\Pi_{NEL}}{B \wp K_B}$$

If $\frac{\perp}{(A \triangleleft B) \wp C} \Pi_{NEL}$ then there exist K_A, K_B such that

$$\frac{K_A \triangleleft K_B}{K} \Pi_{NEL}, \quad \frac{\Pi_{NEL}}{A \wp K_A}, \quad \frac{\Pi_{NEL}}{B \wp K_B}$$

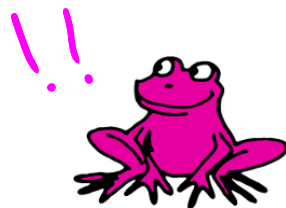
If $\frac{\perp}{!A \wp C} \Pi_{NEL}$ then there exist $K_1, \dots, K_n$ such that

$$\frac{?K_1 \wp \dots \wp ?K_n}{C} \Pi_{NEL}, \quad \frac{\Pi_{NEL}}{A \wp K_1 \wp \dots \wp K_n}$$

If $\frac{\perp}{?A \wp C} \Pi_{NEL}$ then there exists K such that

$$\frac{!K}{C} \Pi_{NEL}, \quad \frac{\Pi_{NEL}}{A \wp K}$$

If $\frac{\Pi_{NEL}}{a \wp C}$ then $\frac{a^\perp}{C} \Pi_{NEL}$



This looks like the theorem we want, but it cannot be proven

NEL

$$\begin{array}{cccccccc}
\text{ax} \frac{1}{a \wp a^\perp} & \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} & \text{q} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} & \text{w} \frac{\perp}{?A} & \text{g} \frac{??A}{?A} & \text{b} \frac{?A \wp A}{?A} & \text{p} \frac{!(A \wp B)}{!A \wp ?B} & \text{e} \frac{1}{!1}
\end{array}$$

NEL

(i) $\frac{\Phi' \Pi}{\text{w} \frac{\perp}{?A} \wp C}$

(ii) $\frac{\Phi' \Pi}{\text{b} \frac{?A \wp A}{?A} \wp C}$

(iii) $\frac{\Phi' \Pi}{\text{g} \frac{??A}{?A} \wp C}$

$$\begin{array}{c}
!K \\
\parallel \\
C, A \wp K
\end{array}$$

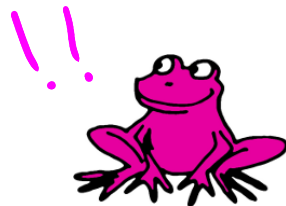
IH not satisfied

Breaks previous cases

If $\frac{\Phi' \Pi_{\text{NEL}}}{?A \wp C}$ then there exists K such that

$$\begin{array}{c}
!K \\
\parallel_{\text{NEL}} \\
C, A \wp K
\end{array}$$

Proof is by induction on
 $(|(A \otimes B) \wp C|, n(\Phi))$



$$\begin{array}{c}
 \text{ax} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \downarrow \frac{\perp}{?A} \quad \text{g} \downarrow \frac{??A}{?A} \quad \text{b} \downarrow \frac{?A \wp A}{?A} \quad \text{p} \downarrow \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \downarrow \frac{1}{!1}
 \end{array}$$

If $\frac{\perp}{(A \otimes B) \wp C} \Pi_{NELC}$ then there exist K_A, K_B such that

$$\frac{K_A \wp K_B}{K} \parallel_{NELC}, \quad \frac{\Pi_{NELC}}{A \wp K_A}, \quad \frac{\Pi_{NELC}}{B \wp K_B}$$

If $\frac{\perp}{(A \triangleleft B) \wp C} \Pi_{NELC}$ then there exist K_A, K_B such that

$$\frac{K_A \triangleleft K_B}{K} \parallel_{NELC}, \quad \frac{\Pi_{NELC}}{A \wp K_A}, \quad \frac{\Pi_{NELC}}{B \wp K_B}$$

If $\frac{\perp}{!A \wp C} \Pi_{NELC}$ then there exist $K_1, \dots, K_n$ such that

$$\frac{?K_1 \wp \dots \wp ?K_n}{C} \parallel_{NELC}, \quad \frac{\Pi_{NELC}}{A \wp K_1 \wp \dots \wp K_n}$$

If $\frac{\perp}{?A \wp C} \Pi_{NELC}$ then there exists K such that

$$\frac{!K}{C} \parallel_{NELC}, \quad \frac{\Pi_{NELC}}{A \wp K}$$

If $\frac{\Pi_{NELC}}{a \wp C}$ then $\frac{a^\perp}{C} \parallel_{NELC}$

If we only have the 'core' rules then the splitting theorem is provable

NELC

$$\begin{array}{c}
 \text{aif} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \downarrow \frac{\perp}{?A} \quad \text{g} \downarrow \frac{??A}{?A} \quad \text{b} \downarrow \frac{?A \wp A}{?A} \quad \text{p} \downarrow \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \downarrow \frac{1}{!1}
 \end{array}$$

NELC

If $\begin{array}{c} A \\ \parallel_{\text{NEL}} \\ B \end{array}$ then $\begin{array}{c} A \\ \parallel_{\text{w}\uparrow, \text{g}\uparrow, \text{b}\uparrow} \\ A' \\ \parallel_{\text{NELC}} \\ B' \\ \parallel_{\text{w}\downarrow, \text{g}\downarrow, \text{b}\downarrow} \\ B \end{array}$

Derivations can be
 decomposed into
 phases



$$\begin{array}{c}
 \text{ail} \frac{1}{a \wp a^\perp} \quad \text{s} \frac{A \otimes (B \wp C)}{(A \otimes B) \wp C} \quad \text{q} \frac{(A \wp B) \triangleleft (C \wp D)}{(A \triangleleft C) \wp (B \triangleleft D)} \quad \text{w} \downarrow \frac{\perp}{?A} \quad \text{g} \downarrow \frac{??A}{?A} \quad \text{b} \downarrow \frac{?A \wp A}{?A} \quad \text{p} \downarrow \frac{!(A \wp B)}{!A \wp ?B} \quad \text{e} \downarrow \frac{1}{!1}
 \end{array}$$

NELC

If $\begin{array}{c} A \\ \parallel_{\text{NEL}} \\ B \end{array}$

then

$\begin{array}{c} A \\ \parallel_{\text{w}\uparrow, \text{g}\uparrow, \text{b}\uparrow} \\ A' \\ \parallel_{\text{NELC}} \\ B' \\ \parallel_{\text{w}\downarrow, \text{g}\downarrow, \text{b}\downarrow} \\ B \end{array}$

Derivations can be
decomposed into
phases

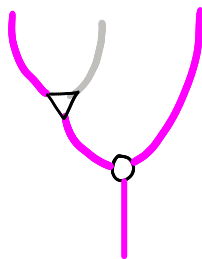


$$\begin{array}{c}
 \text{b} \downarrow \frac{?A \wp A}{?A} \otimes !B \\
 \text{p} \uparrow \frac{\quad}{?(A \otimes B)}
 \end{array}$$



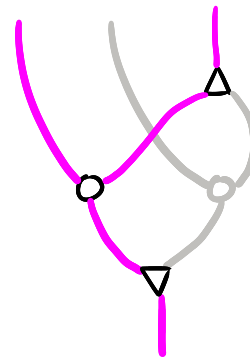
$$\begin{array}{c}
 \text{zs} \frac{(?A \wp A) \otimes \text{b} \uparrow \frac{!B}{!B \otimes B}}{\quad} \\
 \text{p} \uparrow \frac{?A \otimes !B}{?(A \otimes B)} \wp (A \otimes B) \\
 \text{b} \downarrow \frac{\quad}{?(A \otimes B)}
 \end{array}$$

$$\begin{array}{c}
 \text{?}A \otimes A \\
 \hline
 \text{?}A \\
 \hline
 \text{?}(A \otimes B)
 \end{array}
 \otimes \text{!}B$$

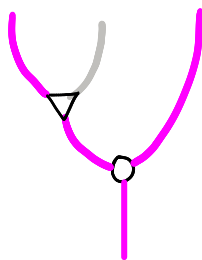


→

$$\begin{array}{c}
 (\text{?}A \otimes A) \otimes \text{!}B \\
 \hline
 \text{?}A \otimes \text{!}B \\
 \hline
 \text{?}(A \otimes B) \otimes (A \otimes B) \\
 \hline
 \text{?}(A \otimes B)
 \end{array}$$

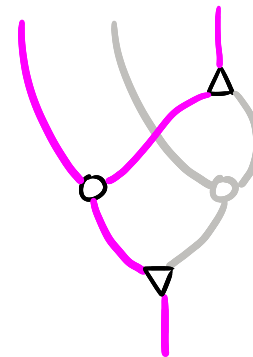


$$\frac{b\downarrow \frac{?A \wp A}{?A} \otimes !B}{p\uparrow ?(A \otimes B)}$$

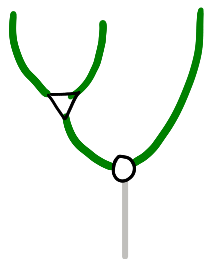


→

$$\frac{2s \frac{p\uparrow \frac{?A \otimes !B}{?(A \otimes B)} \wp (A \otimes B)}{b\downarrow ?(A \otimes B)} \otimes b\uparrow \frac{!B}{!B \otimes B}}$$

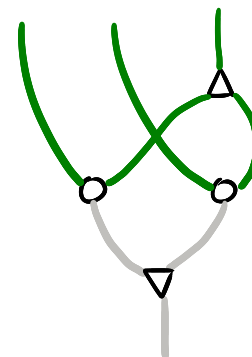


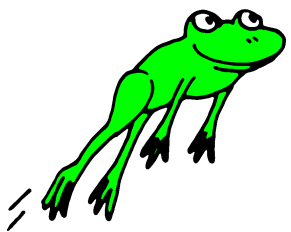
$$\frac{ac\downarrow \frac{a \vee a}{a} \wedge a^\perp}{ai\uparrow \perp}$$



→

$$\frac{2s \frac{ai\uparrow \frac{a \wedge a^\perp}{\perp} \vee ai\uparrow \frac{a \wedge a^\perp}{\perp}}{= \text{or } c\downarrow \perp} \wedge ac\uparrow \frac{a^\perp}{a^\perp \wedge a^\perp}}$$



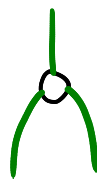


Regularity of inference rules and normalisation procedures



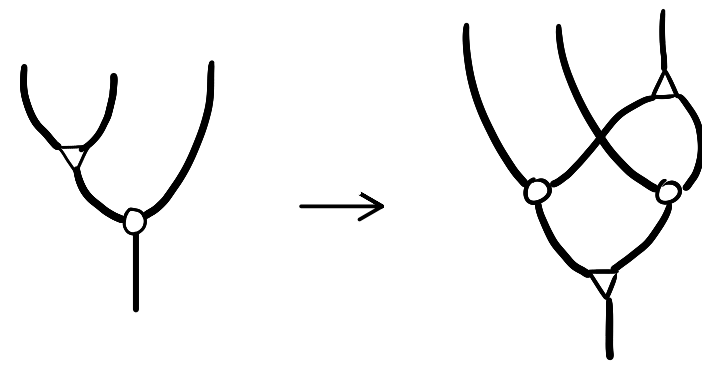
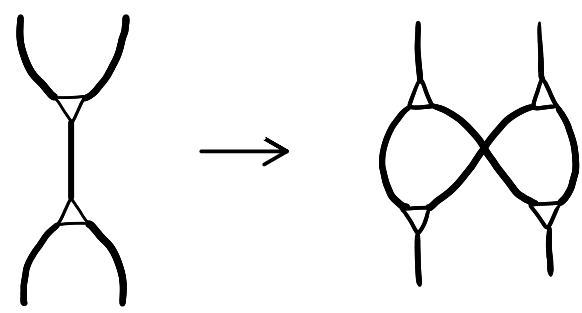
contraction-like rules

$$m_{\alpha}^{\beta} \frac{(A \alpha B) \beta (C \alpha D)}{(A \beta C) \alpha (B \beta D)}$$



identity-like rules

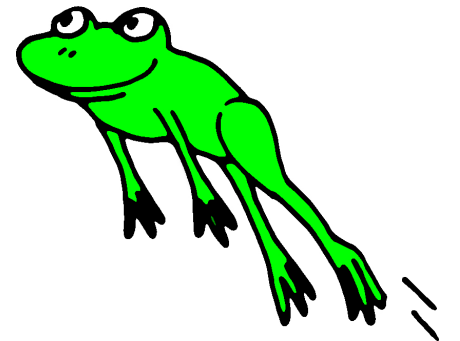
$$s_{\alpha}^{\beta} \frac{(A \alpha B) \beta^{\wedge} (C \alpha D)}{(A \beta^{\wedge} C) \alpha (B \beta^{\vee} D)}$$



Summary

We introduced open deduction and saw how symmetry, atomicity, and low bureaucracy lead to good normalisation

Things we missed : proof compression mechanisms, combinatorial proofs, subatomic logic, ...



References

Linear Logic and Noncommutativity in the Calculus of Structures, Lutz Straßburger's PhD thesis, 2003

A system of Interaction and Structure, ACM Transactions on Computational Logic 8 (1:1) 2007, pp. 1–64,
Alessio Guglielmi

Deep Inference and Symmetry in Classical Proofs, Kai Brännler's PhD thesis, 2004

A General View of Normalisation Through Atomic Flows, Tom Gundersen's PhD thesis, 2010

ESSLLI 2009 course notes, Andrea Aler Tubella and Lutz Straßburger,
available at <https://www.lix.polytechnique.fr/~lutz/orgs/ESSLLI2019-course.html>

Deep inference webpage: alessio.guglielmi.name/res/cos/

